2024-2025 AIAA DBF COMPETITION

TEAM KOREA

SEOUL NATIONAL UNIVERSITY
KONKUK UNIVERSITY
GYEONGSANG NATIONAL UNIVERSITY



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Acronyms, Abbreviations, and Symbols

Symbol	Meaning	Symbol	Meaning
X-1	X-1 test vehicle	M_1 , M_2 , M_3	Mission 1,2,3
$MTOW$	Maximum Take Off Weight	V_{level}	level flight speed
V_{takeoff}	takeoff speed	L/D	Lift over Drag ratio
AR	Aspect Ratio	δ	Oswald efficiency
$C_{m,q}$	Pitching moment damping coefficient	α, AoA	Angle of Attack
C_L, C_{L_t}	lift coefficient, lift coefficient at turn	ρ	density
$C_{L_{max},flap}$	max lift coefficient (flaps deployed)	S, A, S_{ref}	reference area
W_{empty}	empty weight	W_{fuel}	fuel weight
$W_{\text{X-1}}$	X-1 weight	t_{lap}	lap time
t_{level}	level flight time	t_{turn}	turn time
R_{sink}	sink rate	R	turn radius
W	weight	n_{t}	turn load factor
ϕ	bank angle	θ	pitch angle
γ	climb angle, flight path angle	$a_{\text{centripetal}}$	centripetal acceleration
$a_{\text{tangential}}$	tangential acceleration	a_x, a_z	x-axis, z-axis acceleration
T	thrust	TWR	thrust to weight ratio
N_{laps}	Number of laps	Obj_2, Obj_3	Objective 2, Objective 3
$C_{m,\alpha}$	pitching moment coefficient slope	RC	Radio Control
HoQ	House of Quality	QFD	Quality Function Deployment
AHP	Analytic Hierarchy Process	$\lambda_{\max}$	maximum eigenvalue
KV	motor velocity constant	n	factor count at AHP
Q_m	motor torque	ESC	Electronic Speed Controller
ω_n	natural frequency	ζ	damping ratio
CFRP	Carbon Fiber Reinforced Polymer	PA-CF	Polyamide-Carbon Fiber
PLA	Polylactic Acid	LW-PLA	Lightweight Polylactic Acid
FRP	Fiber Reinforced Plastic	GFRP	Glass Fiber Reinforced Polymer
CG	Center of Gravity	NiMH	Nickel-Metal Hydride
NiCad	Nickel–Cadmium	C_D, $C_{D,i}$	Drag, Induced drag coefficient
LiPo	Lithium-Polymer	FORCAST	Flight Optimized RC Aircraft Sizing Tool
$C_{m,0}$	pitching moment coeff at 0 AOA	$C_{n,\beta}$	yawing moment coefficient derivative w.r.t sideslip angle
$C_{l,\beta}$	rolling moment coefficient derivative w.r.t sideslip angle		
$C_{n,T}$	yawing moment coefficient derivative w.r.t thrust		

1 | Executive Summary

This report documents the design, manufacturing, and testing of aircraft *Beomgorae* and X-1 test vehicle "Gaori," developed by Team Korea for the 2024-2025 AIAA Design/Build/Fly Competition. The team's objective is to design an aircraft that maximizes the score of the X-1 Supersonic Flight program, which is composed of three flight missions and one ground mission.

Large payload capacity, alongside high maximum speed and range were design priorities to maximize scores. The aircraft's low-wing configuration and twin-boom mounted H-tail can withstand high load factors, increasing the payload capacity. Furthermore, a dual-motor propulsion system was selected for its superior power efficiency and maximum thrust, increasing maximum speed and range. For the X-1, a flying-wing glider configuration was used to minimize weight while maintaining stability.

The aircraft and X-1 design, manufacturing, and testing were conducted by a team of 37 students, divided into aircraft and X-1 subteams. The design of the aircraft progressed through conceptual, preliminary, and detailed design stages. During conceptual design, trade studies were conducted to determine the initial design of the aircraft. The preliminary design optimized the aircraft main wing span, AR, taper ratio, and twist using FORCAST, a custom built in-house sizing tool. Finally, detail design polished the final designs ready for manufacturing, focusing more on real-world aspects such as manufacturability or structural stability.

Through rigorous analysis, testing, and validation, both "Beomgorae" and "Gaori" demonstrate the capability to successfully complete all mission requirements while maintaining high performance standards, as shown in Table 1.1. The design choices and optimization processes outlined in this report reflect Team Korea's systematic approach to maximizing competition performance while ensuring operational reliability.



Figure 1.1: **Beomgorae, Gaori**

Table 1.1: **Predicted Mission performance**

	M1	M2	M3	GM
Takeoff Weight (kg)	4.15	10	4.58	-
Max Speed (m/s)	32.05	32.52	22.56	-
Mission Time (s)	89.82	118.64	273.18	30
Number of Laps	3	3	8	-

2 | Management Summary

2.1 | Team Organization

Team Korea is a collaboration between the aerospace clubs of Seoul National University (SNU), Gyeongsang National University (GNU), and Konkuk University (KU). Each club is student-led and composed of both undergraduate and graduate students. The team organization is outlined in Figure 2.1.

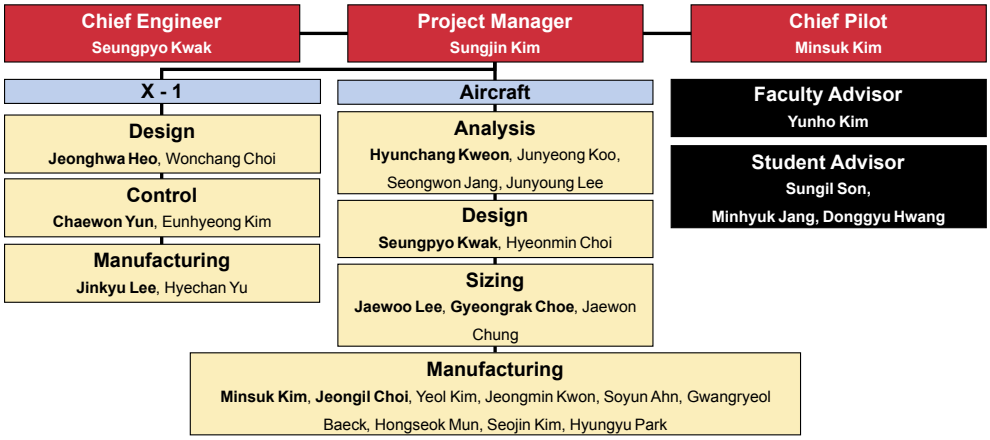


Figure 2.1: Team organization

Faculty advisor provides guidance and supervises the entire process, and graduate students serve as student advisors. The chief engineer along with the project manager leads the entire team. Chief engineer manages the Design Team and oversees overall aircraft design, while the project manager monitors progress and establishes guidelines based on the development timeline.

Each team has a team leader responsible for overseeing overall schedules. While each team had clearly defined roles, close collaboration was emphasized to facilitate seamless cooperation throughout the project. The specific roles assigned to each team are listed in Table 2.1.

Table 2.1: Team Roles

Team		Roles
Aircraft	Analysis	Aerodynamic, Structure, Flight stability analysis
	Design	CAD of all aircraft components
	Sizing	Sizing tool develop, Optimization, Mission Analysis
	Manufacturing	3D printing of aircraft components, Manufacturing aircraft
X-1	Design	CAD of X-1 components, Design of X-1 release mechanism
	Control	Development of autopilot algorithm
	Manufacturing	3D printing of aircraft components, Electric circuit design, Manufacturing X-1

2.2 | Milestone Chart

At the start of the project, a milestone chart was created to establish goals and clearly define team roles. The team leaders, project manager, and chief engineer held weekly meetings to assign tasks for each team

and adjust the schedule. Following this, each team leader conducted weekly meetings within their respective teams to execute the assigned tasks. Table 2.2 shows the milestones and project status up to report submission.

Table 2.2: DBF Milestone Chart

AIAA DBF 2024-2025	September					October				November				December					January				February				March					April			
	01	08	15	22	29	06	13	20	27	03	10	17	24	01	08	15	22	29	05	12	19	26	02	09	16	23	02	09	16	23	30	06	13	20	27
Basic Design																																			
Sensitivity Analysis																																			
Conceptual Design																																			
Optimization Process (FORCAST)																																			
Aerodynamic & Weight Analysis																																			
Mission Analysis																																			
Propulsion Analysis																																			
Algorithm Development and Data Analysis																																			
Final Design																																			
Detailed Design																																			
Design Review																																			
Manufacturing																																			
Components Development																																			
Prototype 1 (Fully 3D printed)																																			
Prototype 2 (Carbon Fiber, Initial configuration)																																			
Prototype 3 (Carbon Fiber, Optimized configuration)																																			
Competition Aircraft																																			
X-1 Prototype																																			
X-1 Final Model																																			
Testing																																			
Preliminary Design Test																																			
Ground Test																																			
Flight Test																																			
Proposal																																			
Report																																			
AIAA DBF Competition																																			

Planned Task
Planned Sub-Task
Actual Progress
Key Dates

3 | Conceptual Design

3.1 | Mission Requirements

The 2024-2025 DBF Competition challenges teams to design and build an aircraft for the X-1 Supersonic Flight Program. Teams are scored based on their reports, mission performance, and participation, with the Total Mission Score comprising three flight missions (M_1, M_2, M_3) and a Ground Mission (GM).

$$\text{Total Mission Score} = \mathrm{M}_1 + \mathrm{M}_2 + \mathrm{M}_3 + \mathrm{GM} \tag{3.1}$$

3.1.1 | Mission Flight Rules

The aircraft must complete a ground rolling takeoff, fly 1000 ft straight sections with 180° bank turns and a 360° turn, and land successfully for scoring. There are no restrictions on turn radius or takeoff distance.

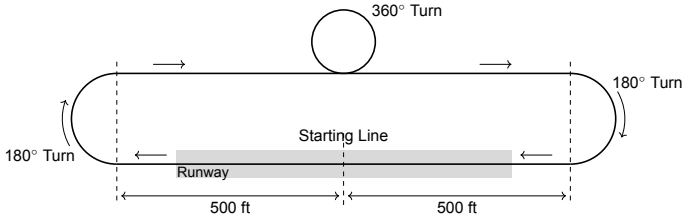


Figure 3.1: Course Layout

Mission 1 – Delivery Flight

Mission 1 requires completing three laps within five minutes for a score of 1.0.

$$\mathrm{M}_1 = \begin{cases} 0.0 \text{ (Unsuccessful Mission 1)} \\ 1.0 \text{ (Successful Mission 1)} \end{cases} \tag{3.2}$$

Mission 2 – Captive Carry Flight

Mission 2 requires carrying filled fuel tanks and X-1 while completing three laps within 5 minutes. Score calculation depends on fuel-weight-to-time ratio.

$$\mathrm{M}_2 = 1 + \frac{\left(\frac{\text{fuel weight}}{\text{time}}\right)_{\text{N}}}{\left(\frac{\text{fuel weight}}{\text{time}}\right)_{\text{MAX}}} \tag{3.3}$$

Mission 3 – Launch Flight

Mission 3 requires the aircraft to take off with the X-1 and empty fuel tanks. The aircraft must complete maximum laps before releasing the X-1 within five minutes. The X-1 must land in the designated bonus box with functioning lights/strobes.

$$\mathrm{M}_3 = 2 + \frac{\left(N_{\text{laps}} + \frac{\text{bonus box score}}{\text{X-1 test vehicle weight}}\right)_{\text{N}}}{\left(N_{\text{laps}} + \frac{\text{bonus box score}}{\text{X-1 test vehicle weight}}\right)_{\text{MAX}}} \tag{3.4}$$

Ground Mission – X-1 Flight Test Program Demonstration

The Ground Mission evaluates team performance and the aircraft's ease of maintenance through a timed demonstration of the X-1 Flight Test Program. Each team must complete aircraft conversion by installing pylons, fuel tanks, the release mechanism and the X-1, and perform a ground launch test, while the aircraft remains standing on the ground (or on blocks).

$$\mathrm{GM} = \frac{(\text{mission time})_{\text{MIN}}}{(\text{mission time})_{\text{N}}} \tag{3.5}$$

3.1.2 | Mission Constraints

Table 3.1 summarizes the key competition constraints that were critically considered in the design process.

Table 3.1: Mission Constraints

Category	Num	Constraints
Configuration	1	Span must not exceed 6 ft (1.82 m).
	2	Assembly must be done while the aircraft is on its landing gear in an upright configuration.
	3	Blocks under landing gear must not exceed fuselage height from the ground.
Structure	4	Aircraft must pass a wing tip load test at MTOW.

Continued on next page

Table 3.1: Mission Constraints (Continued)

Fuel tank	5	Fuel tank must be a commercially available beverage bottle with a screw-on cap, and must not be made of hazardous material.
	6	Bottles must have a minimum capacity of 16 fl. oz (473 mL).
	7	Pylons for external fuel tanks must be removable.
X-1	8	Weight should not exceed 0.55 lbs (249 g).
	9	Flashing lights or strobes should activate after release.
	10	Should be located between external fuel tanks, with a minimum clearance of 0.25 inches (6.35 mm) from aircraft surfaces.
Propulsion	11	A separate battery is required for the Rx/Servo.
	12	Total battery capacity should not exceed 100 Wh.
	13	Arming fuse current rating must not exceed 100 amps

3.1.3 | Problem Statement

Based on the constraints and mission requirements outlined in Table 1, the score factors and key considerations for each mission are summarized in Table 3.2

Table 3.2: Problem Statement

Mission	Score Factors	Constraints
M1	• Stable flight	• Ensure stability and maneuverability for 3 laps within 5 minutes.
M2	• Maximize W_{fuel} • V_{level}	• Maintain stability with MTOW increase while ensuring sufficient speed • Wing loading must stay within a safe range. • Aircraft must land safely without structural failure at the increased stall speed caused by the high takeoff weight. • Fuel tank and X-1 must remain securely attached under all conditions.
M3	• Maximize N_{laps} • Precise X-1 landing • Minimize $W_{\text{X-1}}$	• Both power consumption and lap time must be considered to maximize N_{laps}. • X-1 must be released at the planned timing and land at the designated location. • X-1 must be lightweight while achieving a high descent rate and a reliable landing. • Upon release, X-1 must activate its LED system, and its circuitry must be robust enough to withstand impact forces.
GM	• Fast and secure flight setup	• Ensure reliable flight control. • X-1 must reliably detect its release and activate its LED system accordingly.

3.2 | Sensitivity Analysis

3.2.1 | Theoretical Approach

To conduct the sensitivity analysis, simplified score equations with multiple variables were derived for each mission. Throughout this process, level flight speed, turn speed, and bank angle during turns were assumed to be constant. Additionally, level flight speed and turn speed were analyzed as independent variables. Figure 3.2 illustrates the forces acting during a coordinated level turn, with the corresponding mathematical relationships presented in equations 3.6 $\sim$ 3.8.

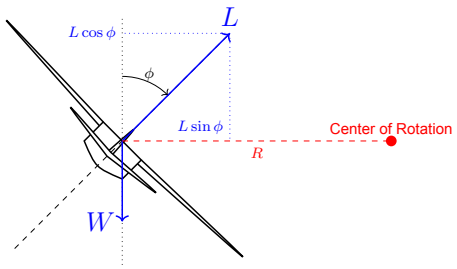


Figure 3.2: Turning Flight

$$\cos\phi = \frac{W}{L} = \frac{1}{n_t} \tag{3.6}$$

$$\frac{1}{\sin\phi} = \frac{n_t}{\sqrt{n_t^2 - 1}} \tag{3.7}$$

$$L\sin\phi = \frac{mV^2}{R} \tag{3.8}$$

Using equations 3.6 $\sim$ 3.8, the turning flight time is derived as follows.

$$R = \frac{mV^2}{L} \cdot \frac{1}{\sin\phi} = \frac{2}{\rho g C_{L_t}} \cdot \frac{W}{S} \cdot \frac{n_t}{\sqrt{n_t^2 - 1}} \tag{3.9}$$

$$V_{\text{turn}} = \sqrt{\frac{2W}{\rho S C_{L_t}}} \cdot \sqrt{n_t} \tag{3.10}$$

$$t_{\text{turn}} = \frac{4\pi R}{V_{\text{turn}}} = \frac{4\pi}{g}\sqrt{\frac{2\left(\frac{W}{S}\right)}{\rho C_{L_t}}} \cdot \sqrt{\frac{n_t}{n_t^2 - 1}} \tag{3.11}$$

In Mission 2, the approximate lap time t_{lap} is calculated by adding the level flight time t_{level} and the turning flight time t_{turn}.

$$W = W_{\text{empty}} + W_{\text{fuel}} + W_{\text{X-1}} \tag{3.12}$$

$$t_{\text{lap}} = t_{\text{level}} + t_{\text{turn}} = \frac{2000}{V_{\text{level}}} + \frac{4\pi}{g}\sqrt{\frac{2W}{\rho S C_{L_t}}} \cdot \sqrt{\frac{n_t}{n_t^2 - 1}} \tag{3.13}$$

The objective value Obj_2 is the fuel weight divided by the time for three laps. In Mission 3, the sink rate of X-1 (R_{sink}) was included as a design parameter. The objective value Obj_3 is the sum of the maximum lap count, added to the bonus score over X-1 weight. The maximum lap count is the floor value of available flight time before release divided by lap time.

$$Obj_2 = W_{\text{fuel}} \;/\; 3t_{\text{lap}} \tag{3.14}$$

$$Obj_3 = \left\lfloor \frac{300 - \frac{200}{R_{\text{sink}}}}{t_{\text{lap}}} \right\rfloor + \frac{\text{Bonus}}{W_{\text{X-1}}} \tag{3.15}$$

Based on the analysis, six variables included in the equations that affect the score are as follows: W_{fuel}, V_{level}, n_t, $W_{\text{X-1}}$, R_{sink} and C_{L_t}.

3.2.2 | Analysis Result

A sensitivity analysis of the mission objective equations revealed V_{level} and $W_{\text{X-1}}$ as the most influential parameters(Figure 3.3). Some plots are discontinuous due to Mission 3's discrete score calculation. Variables that enable more immediate performance gains received higher priority in the design process.

Additionally, analysis of Mission 3 bonus points confirmed that the bonus points are the most influential part of the objective, giving a +24.3% ~ -16.2% score differential compared to the baseline of 1.0 points.

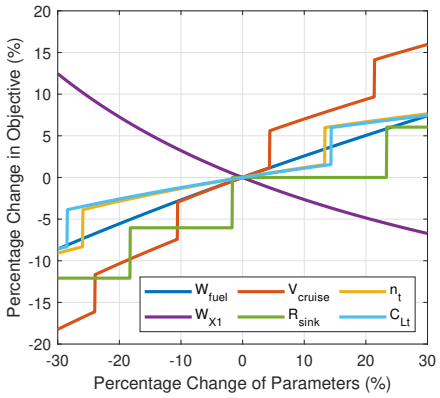


Figure 3.3: **Sensitivity Analysis Result**

3.3 | Subsystem Design Requirements

A trade study was conducted to convert mission requirements into subsystem design requirements. Selection criteria were based on factors for fixed-wing aircraft with empirical significnace. The Analytic Hierarchy Process (AHP) ranked these factors through pairwise comparisons, with results shown in Table 3.3.

Table 3.3: **Result of AHP**

Significantly more important 5 ~ Same importance 1	Flight Stability	Structural Reliability	Payload Capacity	Speed	Range	Controllability	Manufacturability	Transportability	Maintainability	Priority	Rank
Speed	3.00	2.00	3.00	1.00	2.00	4.00	5.00	4.00	5.00	0.252	1
Range	0.50	2.00	3.00	0.50	1.00	4.00	4.00	5.00	5.00	0.177	2
Payload Capacity	3.00	2.00	1.00	0.33	0.33	3.00	4.00	5.00	5.00	0.153	3
Flight Stability	1.00	2.00	0.33	0.33	2.00	3.00	3.00	3.00	4.00	0.133	4
Structural Reliability	0.50	1.00	0.50	0.50	0.50	3.00	4.00	4.00	4.00	0.114	5
Controllability	0.33	0.33	0.33	0.25	0.25	1.00	3.00	4.00	4.00	0.066	6
Manufacturability	0.33	0.25	0.25	0.33	0.25	3.00	1.00	3.00	4.00	0.042	7
Transportability	0.33	0.25	0.25	0.25	0.20	3.00	3.00	1.00	3.00	0.036	8
Maintainability	0.25	0.25	0.20	0.20	0.20	0.33	0.50	0.33	1.00	0.027	9

Judgment consistency was verified using the Consistency Ratio (CR), calculated from the Consistency Index (CI), Random Index ($RI = 1.4$), maximum eigenvalue (λ_{max}), and factor count (n). With $CR = 0.083$ (below the 0.1 threshold), the analysis demonstrated acceptable consistency [1].

$$CR = \frac{CI}{RI} = \frac{\lambda_{\text{max}} - n}{(n-1) \times (1/RI)} \tag{3.16}$$

AHP results established criteria importance rankings that formed the basis for Quality Function Deployment (QFD). Through the House of Quality (HoQ) shown in Figure **??**, QFD translated mission requirements into 13 subsystem design requirements. These requirements were derived from the mission requirements analysis in Section 3.1 and sensitivity analysis results in Section 3.2.

These relationships were scored on a 9-point scale and symbolically represented in Table 3.4. The ranking of subsystem design requirements is presented at the bottom of the table and was considered throughout the overall design process, including the design selection process.

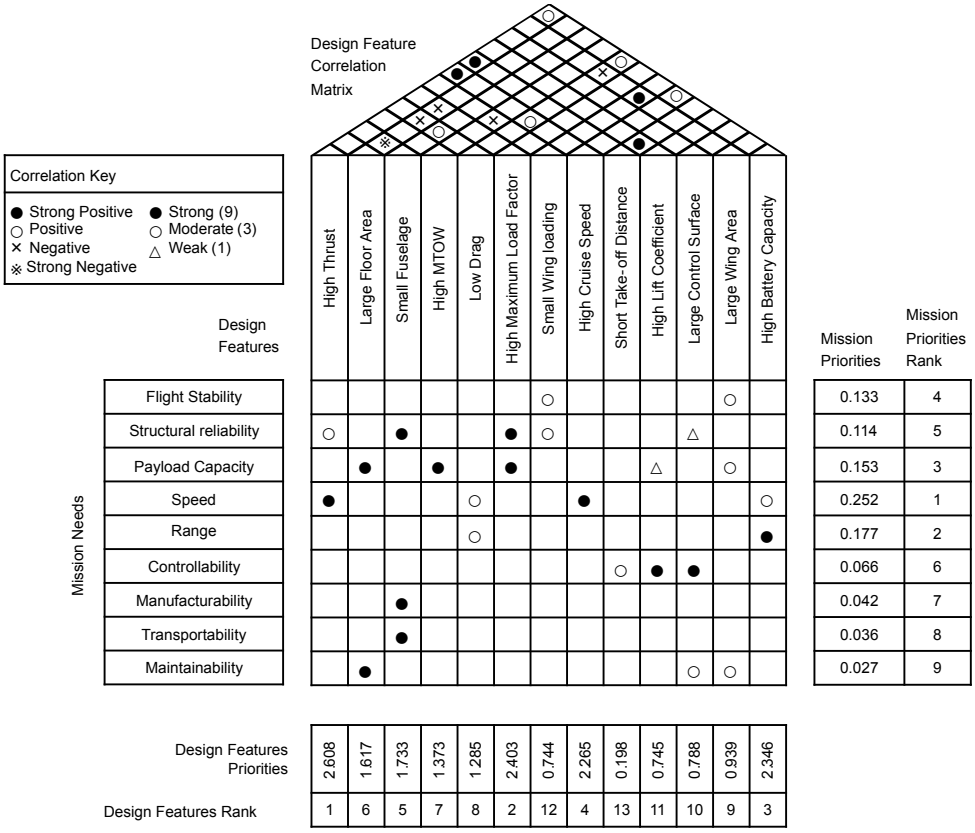


Table 3.4: **House of Quality**

3.4 | Design Selection Process

After summarizing potential concepts and configurations, optimal designs were selected through QFD evaluation. Main Wing, tail Wing, propulsion configurations, payload locations, and landing gear configurations were assessed on a 1-5 scale, weighted by QFD values. The highest-scoring configuration became the final design. QFD weights were established through the HoQ analysis described above.

3.4.1 | Main Wing Configuration

Three main wing configurations were evaluated with QFD analysis prioritizing high load factor, small fuselage, and large mounting space. The **Low-Wing** configuration was selected because it effectively transfers bending stress to the fuselage during lift generation, allowing higher load factors. Also, it provides large mounting space by expanding floor space with wing space for the X-1 release system and landing gear.

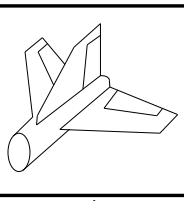
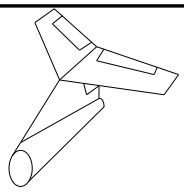
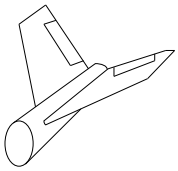
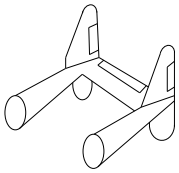
Table 3.5: Mainwing QFD Results

Correlation Key: ● Strong (9), ○ Moderate (3), △ Weak (1), None (0)	Mission Priorities	Flight Stability	Structural Reliability	Payload Capacity	Controllability	Manufacturability	Transportability	Maintainability	QFD Weight	Main Wing Configuration: Low-Wing	Mid-Wing	High-Wing
Priorities	Design Properties											
1.617	Large Mounting Space			●				●	18.195	5	3	4
1.374	High MTOW			●					15.455	5	4	5
2.404	High Maximum Load Factor		●	●					27.044	5	4	3
0.744	Small Wing Loading	○	○						8.370	5	5	4
1.733	Small Fuselage		●			●	●		19.499	5	4	4
0.746	High Lift Coefficient			△	●				8.389	4	5	4
0.789	Large Control Surface		△		●			○	8.873	4	5	4
0.940	Large Wing Area	○		○				○	10.572	5	3	5
	AHP Weight	0.134	0.114	0.153	0.066	0.042	0.036	0.027		564.72	462.45	464.57

3.4.2 | Tail Wing Configuration

Four Tail Wing configurations were evaluated with QFD analysis prioritizing high load factor, large mounting space, and high MTOW. The **Twin-Boom Mounted H-Tail** configuration was selected for its dual carbon booms that provide structural stability and vibration damping for high load factors and high MTOW. This design eliminates the need to extend the fuselage to the tail, significantly reducing weight and drag.

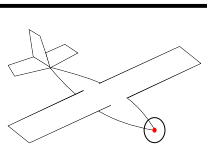
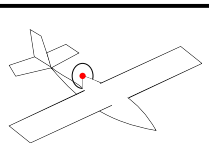
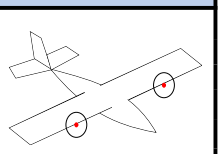
Table 3.6: Tailwing QFD Results

Correlation Key: ● Strong (9), ○ Moderate (3), △ Weak (1), None (0)	Mission Priorities	Flight Stability	Structural Reliability	Payload Capacity	Controllability	Maintainability	Speed	QFD Weight	Tail Wing Configuration: Conventional	T-Tail	V-Tail	Twin-Boom Mounted
Priorities	Design Properties											
1.617	Large Mounting Space			●		●		17.265	4	4	4	5
1.374	High MTOW			●				14.664	4	3	3	5
2.404	High Maximum Load Factor		●	●				25.661	4	3	4	4
0.744	Small Wing Loading	○	○					7.941	4	3	3	5
0.755	Low Drag						○	8.059	4	4	5	5
0.746	High Lift Coefficient			△	●			7.960	5	4	4	4
0.789	Large Control Surface		△		●	○		8.419	4	3	3	5
0.940	Large Wing Area	○		○		○		10.031	5	4	4	5
	AHP Weight	0.134	0.114	0.153	0.066	0.027	0.252		417.99	343.31	377.03	466.38

3.4.3 | Propulsion Motor Configuration

Three motor configurations were evaluated with QFD analysis prioritizing high thrust, high level flight speed, and high MTOW. The **Dual Motor** configuration was selected for its performance advantages. This design utilizes independent circuits for each motor, generating large thrust while maintaining manageable current in each circuit. Dual motor configuration has additional advantage in that it reduces wake interference and minimizes propeller collision risks during the release of X-1 compared to single-motor alternatives, making it particularly suitable for the mission requirements.

Table 3.7: Propulsion QFD result

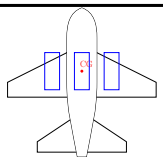
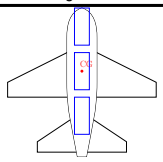
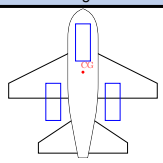
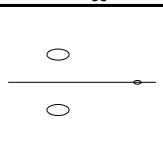
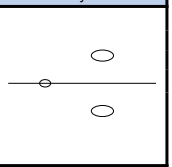
Correlation Key: ● Strong (9), ○ Moderate (3), △ Weak (1), None (0)		Mission Priorities	Flight Stability	Structural Reliability	Payload Capacity	Controllability	Range	Speed	QFD Weight	Motor Configuration: Tractor	Pusher	Dual
Priorities	Design Properties											
1.374	High MTOW				●				18.632	3	3	5
0.198	Short Takeoff Distance					○			2.681	4	4	5
0.744	Small Wing Loading	○	○						10.090	4	5	4
2.608	High Thrust		○					●	35.375	3	3	5
1.591	High Cruise Speed						●		21.579	4	4	5
0.858	Large Wing Area	○		○					11.643	5	4	4
	AHP Weight	0.134	0.114	0.153	0.066	0.177	0.252			357.64	356.08	478.27

3.4.4 | Payload and Landing Gear Configuration

Three payload and two landing gear configurations were evaluated with QFD analysis prioritizing high load factor, large mounting space, and small fuselage. The **Triangular payload mounting** with **Tricycle landing gear** configuration was selected.

The Triangular configuration mounts payloads under wings to counteract lift-induced bending moments, increasing load factor capacity. This arrangement positions fuel tanks near the nose, creating space for the X-1 release mechanism and carbon spar while allowing a more compact fuselage design. With the boom-mounted H-tail, the Tricycle landing gear provides better structural support during takeoff and landing.

Table 3.8: Payload Glider QFD Results

Correlation Key: ● Strong (9), ○ Moderate (3), △ Weak (1), None (0)		Mission Priorities	Flight Stability	Structural Reliability	Payload Capacity	Controllability	Maintainability	Manufacturability	QFD Weight	Payload Mounting Configuration: Lateral	Longitudinal	Triangular	Landing Gear Configuration: Tail Dragger	Tricycle
Priorities	Design Properties													
1.617	Large Mounting Space				●		●		17.076	4	3	5	5	4
1.374	High MTOW				●				14.504	3	3	5	4	5
2.404	High Maximum Load Factor			●	●				25.380	4	3	4	4	4
0.744	Small Wing Loading	○	○						7.855	3	5	4	4	4
0.198	Short Takeoff Distance					○			2.087	4	3	5	4	5
1.406	Small Fuselage			●				●	14.849	5	3	5	3	5
0.789	Large Control Surface			△		●	○		8.327	4	5	3	5	5
0.940	Large Wing Area	○			○		○		9.921	4	5	4	5	5

3.4.5 | Design Selection Process of X-1

Based on sensitivity analysis, the X-1 design prioritized two key objectives: First, achieving reliable landings within the Bonus Box to secure the 2.5 point Bonus Score. Weather data from the competition venue indicated variable conditions from light breezes to greater than 30 mph gusts, necessitating a flight controller despite its added weight.

Second, minimizing overall weight while maintaining necessary functionality. The control system was limited to essential components (flight controller, GPS, and servos) with just two elevons providing both pitch and roll control. Among the tailless configurations considered, the flying wing design was selected for its superior lift-to-weight efficiency by eliminating the fuselage while preserving wing area.

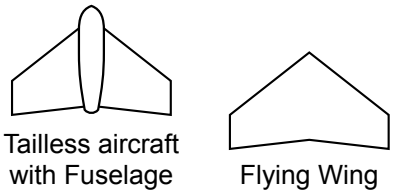


Figure 3.4: **X-1 fuselage configurations**

3.5 | Final Conceptual Design

The final design of the mothership is a low-wing, twin-boom mounted H-tail aircraft with dual motors, a tricycle landing gear, and triangular payload mounting. The X-1 adopts a flying-wing profile with wingtip extensions.

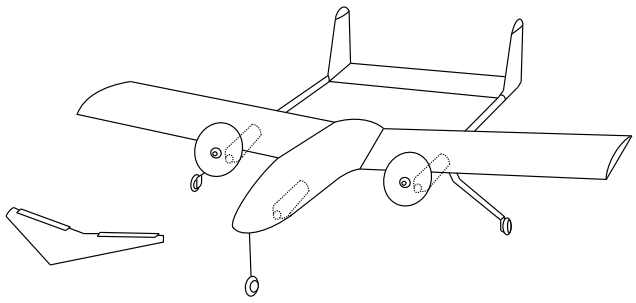


Figure 3.5: **Final Conceptual Design**

4 | Preliminary Design

In the preliminary design phase, we derive mission-optimized parameters based on the conceptual design by performing aircraft flight performance analysis to optimize both configuration and flight strategy.

4.1 | Structure of FORCAST

4.1.1 | Design and Analysis Methodology

Team Korea developed **FORCAST** (Flight Optimized RC Aircraft Sizing Tool), a custom Python-based aircraft sizing tool. Though specifically created for this competition, FORCAST's adaptability makes it valuable for general RC aircraft sizing applications. The tool is available at https://github.com/Bulnabi-SNU/DBF2025_FORCAST. Figure 4.1 illustrates the tool's overall algorithm.

FORCAST employs grid search optimization across a multidimensional parameter space. Design parameters (e.g. wingspan, aspect ratio, and taper ratio) define the aircraft's configuration, while mission parameters (e.g. MTOW and thrust ratios for climb, level flight, and turn) control flight characteristics. As detailed in Section 4.1.2, all parameters vary at predetermined intervals within established ranges. This approach creates a discrete n-dimensional grid where each point represents a unique combination of aircraft configuration and flight strategy.

The sizing tool first validates whether each grid point satisfies predefined constraints such as wing loading. For viable configurations, it performs two key analyses: the aerodynamic analysis module utilizes OpenVSP

and Ansys Fluent to generate lift and drag curves. Meanwhile, the weight analysis utilizes OpenVSP's geometry calculations to determine the aircraft's empty weight (i.e. excluding fuel tanks).

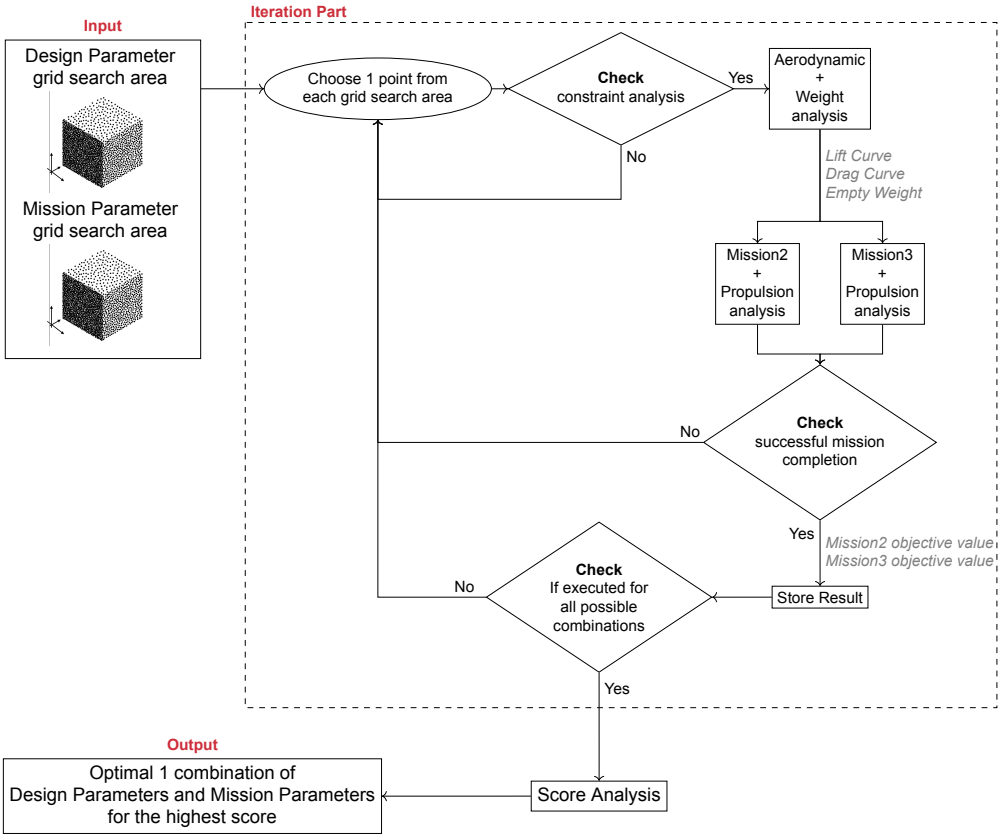


Figure 4.1: Flow Chart of FORCAST

The mission and propulsion analysis conducts flight simulations by solving motion equations at small time intervals for each parameter configuration. This determines flight characteristics including velocity, position, and attitude. The module integrates propulsion analysis to calculate thrust, power consumption, and battery charge based on airspeed and throttle. After the simulation is finished, objective values from Equations 3.14 and 3.15 are recorded with the corresponding mission data.

After processing all configurations in the parameter space, the collected results undergo score analysis to identify the optimal design and mission parameter combination that yields the highest score.

Due to differing computational demands, the analysis was split into two phases: aerodynamic/weight analysis and mission/propulsion analysis. The aerodynamic and weight analysis, which involved analyzing thousands of different design parameter configurations, was carried out on a local high-performance workstation. The mission and propulsion analysis phase, which required processing a significantly larger parameter space, was distributed across a parallel computing infrastructure.

A custom workload distribution system was developed to automatically partition the parameter space and assign computing resources to analyze specific configuration subsets. This parallel processing approach reduced computation time from several weeks of sequential processing to approximately 24 hours. Analysis results were analyzed to determine the optimal configuration that maximized the overall mission score. Explanations of each analysis block are provided in Section 4.2 to Section 4.4.

4.1.2 | Design and Mission Parameters Selection

These parameters were selected as design variables due to their significant performance trade-offs, with justifications provided in Table 4.1. The parameter ranges and intervals, along with fixed parameter values, are described in Section 4.2.1.

Table 4.1: Design Parameter Selection

Parameter	Effects		Optimization
	Increase	Decrease	
Span	Increase in maximum wing area Increase in lift Decrease in wing loading	Reduction in drag Reduction in weight	O
Aspect Ratio	Reduction in induced drag	Increase in wing area	O
Taper Ratio	Increase in structural stability at the wingtips	Reduction in induced drag due to elliptical lift distribution	O
Twist	Delay in wingtip stall Generates elliptical lift distribution	Increase in maximum lift	O
Airfoil	Lift curve, drag curve, and L/D influence the mission flight configuration. As discrete variables, several airfoils are selected for comparison and evaluation		O
Sweep Angle	Due to manufacturing constraints and the effectiveness of zero sweepback at low Reynolds numbers, fixed at 0°		X
Dihedral Angle	With no sweep on the low-wing configuration, a dihedral angle is set to ensure lateral stability		X
Tail Wing Parameters	With the tail wing configuration determined, the effects of tail design parameter variations are minimal		X

The mission parameters derived from a trade study include MTOW and the thrust ratio for each flight phase, with detailed analysis provided in Table 4.2. Thrust ratio is defined as the ratio of the current thrust to the maximum available thrust at a given moment. The specific minimum, maximum, and interval values for the mission parameters, along with the propulsion system specifications, are described in Sections 4.3.1 and 4.4.

Table 4.2: Mission Parameter Selection

Parameter	Effect		Optimization
	Increase	Decrease	
MTOW	Increase in fuel weight	Improved flight performance due to rapid acceleration and lower stall speed	O
Mission2/3 maximum speed	Decrease lap time due to faster speed	Decrease flight duration due to higher power consumption	O
Mission2/3 climb thrust ratio	Enhanced flight performance and reduced lap time due to increased thrust	Increased flight duration due to lower power consumption	X
Mission2/3 level flight thrust ratio Mission2/3 turn thrust ratio			O

Continued on next page

Table 4.2: Mission Parameter Selection (Continued)

Propeller Specification Battery Specification Motor Specification	The propulsion system is determined through a trade study rather than optimization due to regulatory constraints and commercial availability	X

4.2 | Aerodynamic and Weight Analysis

This chapter summarizes all parameters in the aerodynamic and weight analysis, and determines the grid search range for optimization. For parameters that are not subject to optimization, a fixed value is assigned. Explanations of the aerodynamic analysis and weight analysis modules within the sizing tool, along with their validation processes, are presented below.

4.2.1 | Analysis Parameter Description

The optimization parameters determined in Table 4.1 are as follows: span, aspect ratio, taper ratio, twist, and airfoil.

The five optimization parameters are summarized in Table 4.3. The total number of grid points is calculated as 5 (span) × 9 (AR) × 11 (taper) × 3 (twist) × 2 (airfoil) = 2,970, leading the sizing tool to analyze 2,970 different aircraft design configurations.

Table 4.3: Optimization Parameter Grid

	Min	Max	Interval
Span (mm)	1700	1800	25
AR	4	6	0.25
Taper Ratio	0.4	0.9	0.05
Twist (degree)	0	2	1
Airfoil	SG6043 / E216		

Span

The maximum span was set to 1.80 m, since regulations limit the span to 6 ft (1.83 m). The analysis in Table 3.4 identified high MTOW and large wing area as subsystem design requirements. Therefore, the minimum span was set relatively long at 1.70 m.

Aspect Ratio

For a typical homebuilt aircraft, the possible aspect ratio ranges from 4.0 to 12.0 [2]. Since the maximum span is constrained, achieving a large wing area requires a lower aspect ratio. Therefore, the analysis was conducted near the lower end of the feasible range - between 4.0 and 6.0.

Taper Ratio

As analyzed in Table 4.1, taper ratio involves a trade-off between induced drag and structural stability. To assess its effect on induced drag, the relationship between Oswald efficiency and taper ratio was examined [3]. Within the predetermined AR range of 4.0 to 6.0, induced drag increases sharply when the taper ratio falls bellow 0.2, but increases minimally with higher ratios. Considering that a lower taper ratio increases structural instability, the range was set between 0.4 and 0.9.

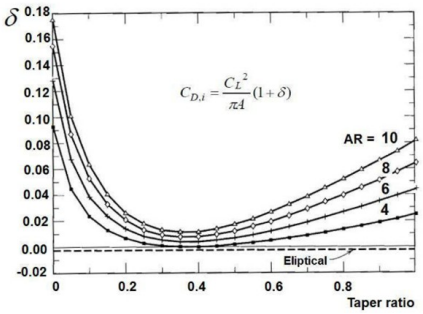


Figure 4.2: Oswald efficiency vs taper ratio for different AR [3]

Twist

Twist (washout) angle typically ranges between 0° and 4° [4]. Excessive twist reduces lift, so the analysis was conducted within the range of 0° to 2°.

Airfoil

Airfoil candidates were selected based on 2D aerodynamic analysis using XFLR5 [5]. Analysis was conducted on 1,648 airfoils from the UIUC Airfoil Coordinates Database [6] at Reynolds number of 700,000 and Mach number of 0.1, based on the normal flight speed and conditions in Tucson during April. The priority for airfoil selection is based on achieving a high lift coefficient, high level flight speed (M2, M3), and a high lift-to-drag ratio for extended range (M3), as determined by the subsystem design requirements analyses.

To determine whether or not a symmetric airfoil would be used, the symmetric airfoil S9027, which features a high stall angle and a relatively high lift-to-drag ratio was considered. However, it was eliminated as it did not meet the high lift coefficient requirement from the subsystem design requirements (Figure 4.3(a)).

Cambered airfoils must operate at a lower angle of attack during high-speed level flight. To meet this requirement, SG6043 and E216, which exhibit a high lift-to-drag ratio at low angles of attack, were selected. As shown in Figure 4.3, both airfoils demonstrate higher lift coefficients and lift-to-drag ratios compared to S9027. Therefore, SG6043 and E216 were selected as the final airfoil candidates.

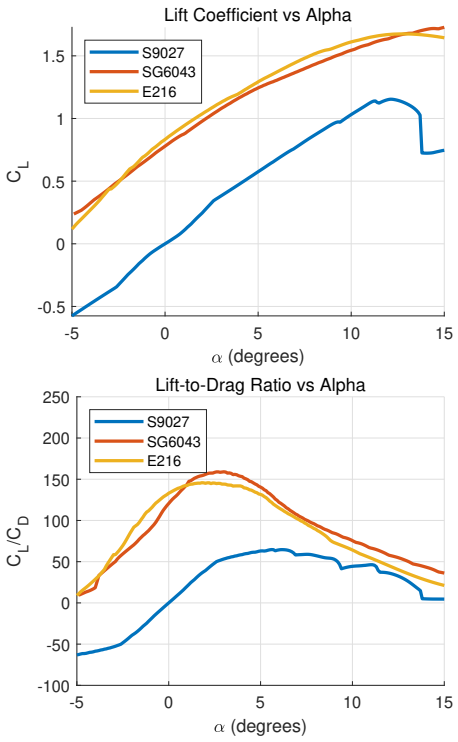


Figure 4.3: (a) Lift curves for three selected airfoils (b) Lift-to-drag ratio curves for the same airfoils

Wingtip

With the aircraft's maximum span limited to 6 ft, a square wingtip was implemented to maximize the wing area within this limit. Also, as shown in Figure 4.4, the square wingtip reduces induced drag by driving the wingtip vortex away from the wing while providing a slight effective span gain, without extra weight or increasing friction drag [7].

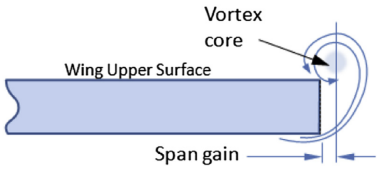


Figure 4.4: General shape of the square wingtip [7]

Sweep, Dihedral Angle

Having no leading edge sweepback eliminates concerns about propeller-wing interference and reduces size of motor mounts. Also, sweepback is aerodynamically inefficient at low Mach numbers and worsens stall characteristics [4]. Therefore, the leading edge sweep angle was set to 0°, resulting in a very low sweepback angle. Given that the design features a low-wing configuration with low sweepback, lateral instability was expected. To compensate this, a 5° dihedral angle was implemented to enhance flight stability. A dihedral angle beyond 5° was considered difficult to manufacture and structurally unstable due to discontinuities in

spar connections.

Tail Design Parameter

The parameters required for the Twin-Boom Mounted H-Tail design, determined during the conceptual design phase, are as follows: volume ratio, area ratio, AR, and taper ratio.

These four parameters fully define the root chord, tip chord, span, and location of the tail. The area ratio and AR of the vertical wing are set by the horizontal wing. The volume ratio must remain within a specific range for flight stability. Therefore, the values were determined based on the configuration of reference RC aircraft with a Twin-Boom Mounted H-Tail [8] . A thin symmetric NACA 4-digit airfoil was selected for its favorable stall characteristics and symmetric control. Unlike the main wing, the tail surfaces are not constrained by span limitations, making the application of wingtips advantageous for reducing induced drag. Therefore, a structurally simple Hoerner wingtip was implemented on the vertical stabilizers.

Control Surface

The control surfaces include the elevator, rudder, and aileron. Additionally, a flap is installed to assist with takeoff at the high MTOW required in Mission 2. Three factors were considered in the control surface sizing process: the boom position, the position when the X-1 is mounted, and ensuring that there is no interference between control surfaces during movement. Considering the boom position, the flaps are divided into two sections. The spanwise and chordwise range of the control surfaces are determined based on empirical data [9].

Table 4.4: Non-optimized Wing Parameters

Main Wing		Horizontal Tail Wing		Vertical Tail Wingw	
Leading Edge Sweep Angle	0°	Volume Ratio	0.7	Volume Ratio	0.053
Dihedral Angle	5°	Area Ratio	0.25	Taper Ratio	0.6
Wingtip	Square Wingtip	AR	4.0	Airfoil	NACA 0009
Flap 1 Spanwise Range	18% ~ 34%	Taper Ratio	1.0	Rudder Spanwise Range	10% ~ 90%
Flap 2 Spanwise Range	40% ~ 62%	Airfoil	NACA 0008	Rudder Chordwise length	35%
Flaps Chordwise Length	35%	Wingtip	Hoerner Wingtip		
Aileron Spanwise Range	65%~95%	Elevator Spanwise Range	0% ~ 70%		
Aileron Chordwise Range	30%	Elevator Chordwise Length	45%		

4.2.2 | Aerodynamic Analysis Module

The aerodynamic analysis utilizes OpenVSP to generate parametric 3D models from aerodynamic design parameters [10]. OpenVSP's Python API enables integration with the sizing tool. Aerodynamic analysis is conducted using VSPAero, OpenVSP's aerodynamic analysis tool, which employs the vortex-lattice method (VLM) for fast computations of lifting surfaces. However, VLM has limited accuracy due to its inability to predict stalling or viscous flow. Therefore, VSPAero is used exclusively for the aerodynamic analysis of lifting surfaces, including the main and tail wings. An example of the lifting surfaces geometry generated by OpenVSP is shown in Figure 4.5. VSPAero generates lift and drag curves, requiring the AoA range for analysis. In a normal flight, AoA varies

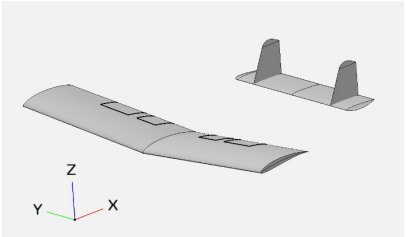


Figure 4.5: Aircraft lifting surface generated in OpenVSP

between -3° and 10°, with stall angle ≈ 13°. Consequently, analysis was performed at 1° intervals from -3° to 10°. For the flaps-deployed cases, the lift and drag coefficients were analyzed only at AoA's of 0° and 10°.

Since drag is generated by all components, it requires further refinement. Given that the fuselage size remains constant throughout the sizing process, its drag was precomputed with Ansys Fluent and added to the VSPAero analysis results. The difference between the total aircraft drag (via Ansys Fluent) and the drag generated by the lifting surfaces (via VSPAero) was taken as the fuselage drag. The analysis with Ansys Fluent are detailed in Section 4.7.1. The resulting fuselage drag coefficient graph is presented in Figure 4.6.

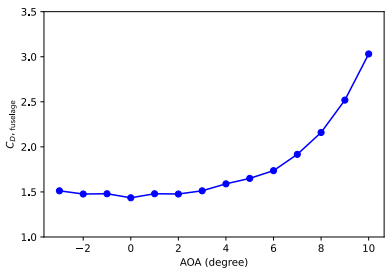


Figure 4.6: Fuselage Drag Coefficient (S_{ref}: Maximum cross-sectional area)

In conclusion, the aerodynamic analysis module outputs the lift and drag curves of the aircraft, along with aerodynamic coefficients at 0° and 10° with flaps deployed.

4.2.3 | Weight Analysis Module

The aircraft's empty weight can be broadly categorized into three main components: fuselage, wing, and boom. To achieve accurate estimation without relying on empirical formulas, the weight analysis parameters were derived using data from the first prototype aircraft. Since the fuselage remains unchanged throughout the sizing process, its weight was directly used. Detailed specifications are provided in Section 5.4.

For the wing, weight estimation considered not only the outer skin but also internal structures such as ribs and spars. Instead of assuming proportionality to surface area, the wing's density was assumed to be constant. The prototype wing's density was used as reference.

Finally, the two booms connecting the main and tail wings were analyzed by its linear density (g/mm). Using the boom length obtained from OpenVSP, the total boom mass was approximated. The reference values are summarized in Table 4.5.

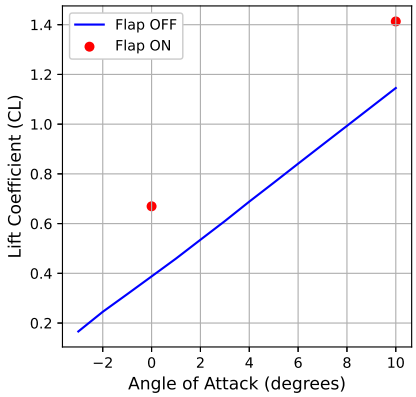
Table 4.5: Weight Analysis Parameters

Fuselage Weight (g)	2500
Wing Geometry Density (g/m^3)	5.88×10^{-5}
Boom Density (g/mm)	0.121

The overall output of the aerodynamic and weight analysis modules combined is presented in Figure 4.7.

Empty Weight (g)	4516.28
Wing Weight (g)	1549.05
Boom Weight (g)	267.23
Fuselage Weight (g)	2500.00
S (m^2)	0.6821

Span (m)	1.80
AR	4.71
Taper	0.90
Incidence (degree)	0.00
Twist (degree)	2.00



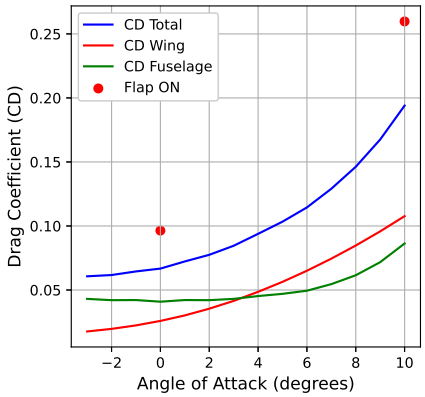


Figure 4.7: Output of Aerodynamic and Weight Analysis Module

4.2.4 | Analysis Module Validation

Evaluating whether the analysis module produces valid results is essential for ensuring the accuracy of the analysis results. For the same design parameters, the generated geometry was analyzed using both Ansys Fluent and the aerodynamic analysis module. The results were then compared at Figure 4.8.

The analysis confirmed that stall prediction is not possible as expected. However, within the operational range of -3° to 10°, the lift and drag coefficients closely follow the trend with a mean square error of 0.048 and 0.014 respectively, demonstrating the reliability of FORCAST's aerodynamic analysis module.

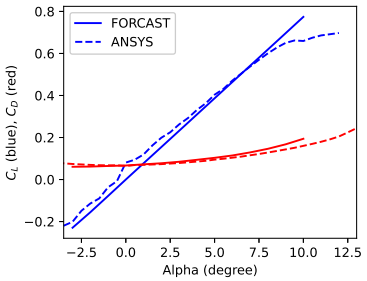


Figure 4.8: Comparison of aerodynamic coefficient

4.3 | Mission

The Mission Module is a simulation designed to predict mission performance in the missions based on analysis results from the Aerodynamic Module. The mission profile is divided into four phases: Takeoff, Climb, Level flight, and Turn. In each phase, the forces acting on the aircraft are analyzed to calculate the acceleration. Velocity and position is determined by acceleration via the Runge-Kutta method.

4.3.1 | Mission Parameter Description

The optimization parameters listed in Table 4.2 are as follows: MTOW, maximum speed, level flight thrust ratio and turn thrust ratio. The process of determining the range for optimization for each parameter is described below.

MTOW

The QFD analysis identified high MTOW as a key design requirement. Since large MTOW could degrade flight performance and incur structural failure during landing, an upper limit of 10 kg and lower limit of 7 kg were set.

Mission maximum speed

For Mission 2, a range of 28 $\sim$ 40 m/s was set since a high maximum speed results in faster lap completion. For Mission 3, a range of 20 $\sim$ 36 m/s was set since energy efficiency is valued. The speed ranges were determined based on previous DBF competition aircraft's flight speeds.

Thrust Ratio

The climb thrust ratio was fixed at 0.9 due to its minimal impact on mission performance. For the Level flight thrust ratio, a range of 0.4 to 0.9 for Mission 2 to prioritize speed, and a range of 0.2 to 0.7 for Mission 3, to focus on efficiency, was chosen. For the Turn thrust ratio, a range of 0.4 to 0.9 for Mission 2 and a range of 0.3 to 0.8 for Mission 3 was chosen.

Capabilities and Uncertainties

To simplify the simulations, several assumptions were made. Only gravity, thrust, lift, and drag were considered. Since the aircraft was modeled as a point mass, only translational motion was considered, while

rotational motion was ignored. Since rotational motion was not considered, roll, pitch, and yaw rates were not taken into account. Thus, it was also assumed that the pilot could achieve the desired attitude as determined in the simulation. Since the simulation was conducted separately for each flight phase, discontinuities exist between phases, which is not very realistic.

However, the integrals of variables - such as flight trajectory and total flight time - remain continuous despite the discontinuous transitions. This effect can be observed in the velocity and position plots of Section 4.7.4.

4.3.2 | Mission Phases

Takeoff Phase

From 0% to 90% of V_{takeoff}, the aircraft maintains a 0° angle of attack with the flaps fully deployed. Between 90% and 100% of V_{takeoff}, the aircraft transitions to the angle of attack that provides the $C_{L_{max},flap}$ with flaps fully deployed, raising the nose of the aircraft. For simplicity, the nose-lifting motion is assumed to be discrete. While the wheels are in contact with the ground, the equations of motion of the aircraft is:

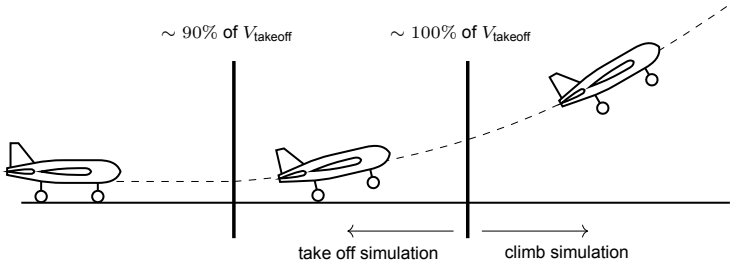


Figure 4.9: Takeoff phase description

$$a_x = \frac{g}{W}\left[T - D - \mu(W - L)\right] \qquad (\mu = 0.03 \text{ @ dry concrete or asphalt}) \tag{4.1}$$

Climb Phase

The flaps are retracted once the aircraft reaches an altitude of 5 m, and is assumed to be instant. During the climb phase, the angle of attack is constrained between -5° and 8°. The maximum load is limited to 30 kg to ensure structural safety and stability. The sideslip angle (β) is assumed to be 0° (i.e. no lateral deviation).

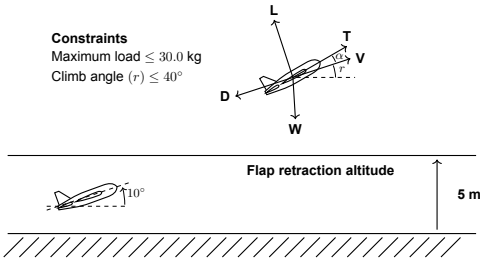


Figure 4.10: Climb phase description

$$a_x = \frac{g}{W}\left[T\cos\theta - L\sin\gamma - D\cos\gamma\right] \tag{4.2}$$

$$a_z = \frac{g}{W}\left[T\sin\theta + L\cos\gamma - D\sin\gamma - W\right] \tag{4.3}$$

Level Flight Phase

When the speed and thrust are specified, the aircraft maintains an AoA (α) that ensures equilibrium between weight (W) and lift (L). If the aircraft reaches the maximum speed limit, the throttle is reduced to balance drag (D) and thrust (T), preventing the speed from exceeding this limit. The aircraft is assumed to maintain a constant altitude during the level flight phase.

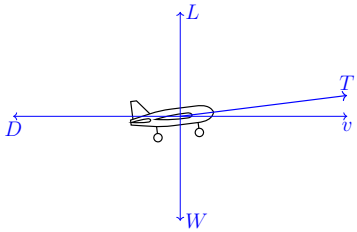


Figure 4.11: Level flight phase description

$$a_x = \frac{g}{W}\left[T\cos\alpha - D\right] \tag{4.4}$$

Turn Phase

The aircraft is assumed to perform a coordinated level turn by tilting the lift vector to provide the required centripetal force while maintaining stability. To ensure constant altitude, the angle of attack is increased so that the net force in the vertical direction is 0. The angle of attack has a maximum limit of 8°.

$$a_{\text{centripetal}} = \frac{g}{W}\left[L \sin \phi\right] \tag{4.5}$$

$$a_{\text{tangential}} = \frac{g}{W}\left[T - D\right] \tag{4.6}$$

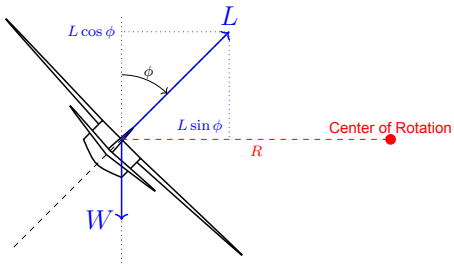


Figure 4.12: Turn phase description

4.3.3 | Overall Mission Evaluation

The missions are constructed by combining multiple mission phases. Furthermore, there are definite termination conditions for each phase.

The takeoff phase ends when the aircraft reaches V_{takeoff}. The climb phase ends when the aircraft reaches the target altitude, but to ensure a smooth transition into level flight, it is additionally required that the flight path angle (γ) reaches 0. Furthermore, to prevent excessive horizontal displacement during the climb, a limit to the x-coordinate was also added. The level flight phase ends when the aircraft reaches the desired x-coordinate. The turn phase ends when the aircraft completes the turn.

An example of a ”mission” is shown below:

Takeoff → Climb → Level flight → Turn → Level flight → Turn → Level flight → Turn → Level flight

Mission 3 requires a high-altitude climb to reach the X-1 release altitude. If the target altitude is not achieved in a single climb, an additional climb phase is introduced after completing a 180° turn. Since Mission 3 does not have a predetermined number of laps, the total mission termination conditions are defined based on the minimum battery voltage and flight duration. The aircraft continues to fly laps for 270 seconds, excluding the 30 seconds required for X-1 landing. However, if the mission termination conditions are met during mid-lap, only completed laps are considered successful.

During the mission simulation completes, parameters such as the number of laps, velocity, position, and time are recorded. Once the mission is complete, the Mission 2 and Mission 3 objectives are calculated and stored as well.

4.4 | Propulsion

The objective of the Propulsion Subsystem design is to find a combination of a motor, propeller, and battery that can operate efficiently in both Mission 2 and 3. Since the objectives of Mission 2 and Mission 3 are different, it is important to ensure balanced performance across both missions.

4.4.1 | Battery Selection

In compliance with regulations, the use of NiCad, NiMH, or lithium-based batteries is permitted. Among these, Lithium-Polymer LiPo batteries have been selected due to their high specific energy and discharge characteristics. The House of Quality (HoQ) analysis revealed that high thrust and high battery capacity are critical design features. To achieve high thrust, a battery voltage of 6S(22.2 V) or higher was considered.

Within the 50 Wh limit per battery due to the twin-motor configuration, the candidates include a 6S pack with a capacity of 2200–2250 mAh (48.84-49.95 Wh) or an 8S pack with a capacity of 1600–1680 mAh (47.36-49.73 Wh).

The primary advantage of an 8S configuration is its higher operating voltage, which reduces current draw. This reduction in current reduces losses and improves the overall efficiency of the propulsion system. However, the availability of motors compatible with this voltage is limited, restricting motor selection for the aircraft. High C-rate was chosen to minimize voltage sag at high discharge currents, ensuring stable power delivery during high throttle operation. Therefore, the **Vega Graphene 2200mAh 6S 70C LiPo battery** was selected.

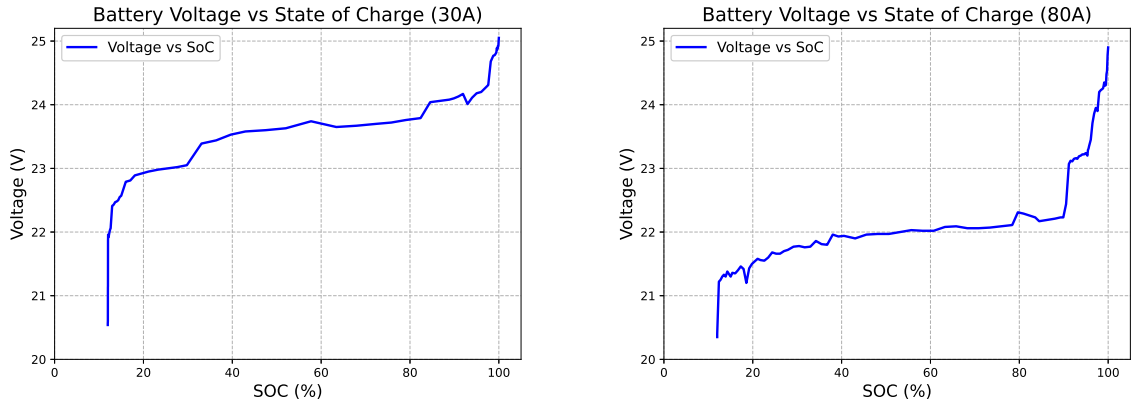


Figure 4.13: Battery discharge data (left: 30A, right: 80A)

To enhance the accuracy of optimization and mission performance prediction, a battery discharge test was conducted. The test was performed at two discharge currents: 30A, corresponding to the expected current at level flight, and 80A, corresponding to the expected current at climb flight. The results indicate that even under an 80A discharge condition, the battery showed relatively stable voltage retention, highlighting its suitability for high-power applications.

4.4.2 | Motor & Propeller Selection

The validity of the motor and propeller combination was assessed through the following process. The table above presents the specifications and performance chart provided by Scorpion for their MII-3011-1100Kv motor[11]. This data was referenced to establish a method for analyzing the motor and propeller combination. Similarly, the internal resistance represents only the motor's resistance, whereas the total resistance of the propulsion circuit is higher.

Therefore, to enhance accuracy, a method was developed to calculate the actual motor specifications based on the performance chart. Rather than using the values provided in the specification sheet, these parameters were directly approximated to better reflect real-world performance. The equations used in the calculation process are as follows [12]. The KV value was calculated by dividing the current (I) by the torque (Q_m), and a conversion constant was multiplied to convert the unit to RPM.

$$\mathrm{KV} = \left(\frac{I}{Q_m}\right) \times \left(\frac{60}{2\pi}\right) \tag{4.7}$$

$$\text{Predicted RPM} = \mathrm{KV} \times (\text{Throttle} \times \text{Voltage} - \text{Current} \times \text{Internal Resistance}) \tag{4.8}$$

Table 4.6: Motor Specification and Performance

Motor Products: Scorpion MII-3011-1100Kv				
Specifications				
Peak Power (W)	Kv (RPM/V)	Peak Current (A)	Weight (g)	Internal Resistance (mΩ)
1332	1100	60	90	55
Performance (Prop size : 8*4*3)				
Throttle (%)	Voltage (V)	Torque (Nm)	Current (A)	RPM
40	24.1	0.06	6.6	10231
50	24	0.102	11.2	12228
60	23.8	0.157	17.3	14114
70	23.6	0.23	25.3	15885
80	23.3	0.321	35.3	17285
90	23	0.425	46.7	18862
100	22.6	0.532	58.5	19862

Based on the table above, the KV value was determined via a linear regression model. The predicted RPM was then compared with the RPM from the data sheet to determine the internal resistance that minimizes the Root Mean Square Error (RMSE). The actual KV of the Scorpion MII-3011-1100KV motor was calculated to be 1049.677 KV with an internal resistance of 0.059 Ω.

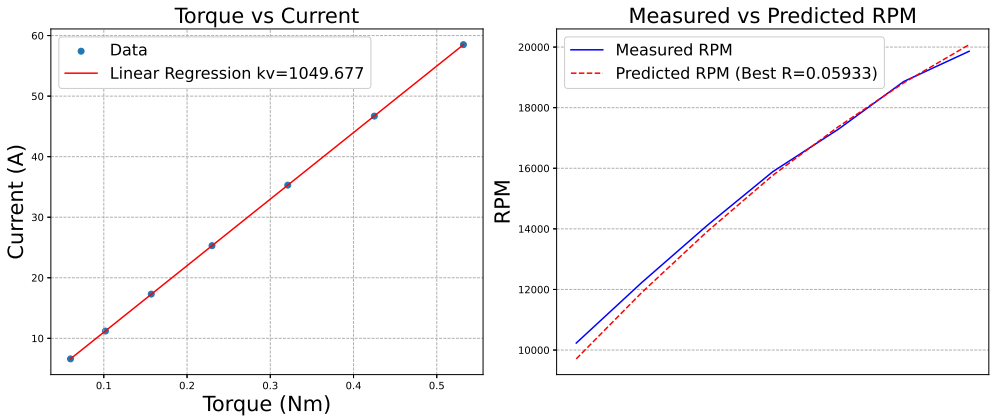


Figure 4.14: Data of Scorpion MII-3011-1100KV

For ease of analysis, only APC propellers were considered, as the APC website provides experimental data for most of its propellers.[13] From this data, the thrust and torque curves of the propeller can be obtained based on RPM and the forward speed of the aircraft. Thus, torque of the motor and propeller can be expressed as a function of RPM.

In the graph below, the motor torque values were calculated using Equation 4.7, while the propeller torque values were directly taken from the experimental data provided by APC. The analysis results correspond to the Scorpion MII-3011-1100KV motor with an APC 10×6E propeller installed.

For example, for the graphs at a forward speed of 20 m/s with varying throttle levels, no intersection points are observed for throttle of 20% and 80%. Since the motor and propeller operate as a coupled system, their RPM must be equal. The absence of an intersection indicates that thrust cannot be generated under these conditions. When throttle is at 20%, the torque generated by the airflow is greater than that provided by the motor, causing windmilling.

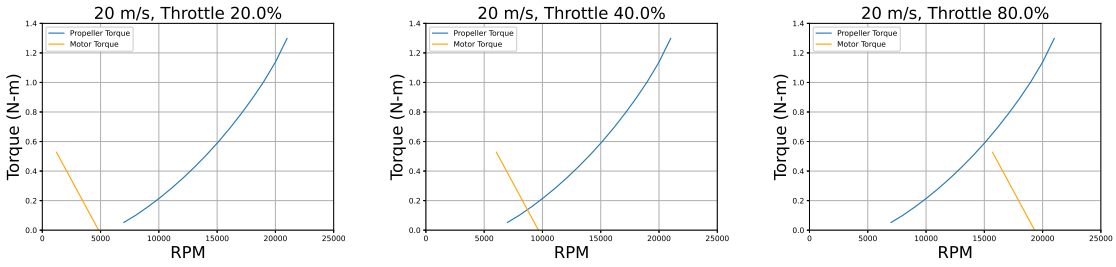


Figure 4.15: **Torque vs RPM at Speed 20 m/s and Throttle Levels**

At 80% thrust, the motor torque curve terminates before an intersection, indicating that the motor has exceeded its maximum current or power limit. In conclusion, it can be determined that using an APC 10×6E propeller with the Scorpion MII-3011-1100KV motor is not suitable. Based on a series of tests evaluating various motor and propeller combinations, three optimal configurations were selected. The selection criteria, based on the HoQ analysis results, are as follows:

1. Power and Current Constraints: The motor must not exceed its maximum current or power limits at any forward velocity up to 40 m/s.
2. Thrust Requirement at High Speed: To achieve a high maximum speed, the thrust at a forward velocity of 40 m/s must be at least 1 kg.
3. Takeoff Performance: The maximum thrust at 0 m/s must be at least 3.5 kg.

These criteria ensure that the selected configurations provide reliable performance across the operational speed range while maintaining current and power limits.

Table 4.7: **Optimal Configurations of Propulsion System**

Configuration	Config 1	Config 2	Config 3
Motor Products	Scorpion MII-3011-1100Kv	Scorpion A-4220-540Kv-V2	Scorpion A-4220-540Kv-V2
Peak Power (W)	1332W	2553W	2553W
Kv (RPM/V)	1100Kv	540Kv	540Kv
Peak Current (A)	60A	115A	115A
Weight (g)	90g	336g	336g
Suggested Propeller	APC 8×6E	APC 15×10E	APC 16×8E
Max LiPo Voltage	6S	6S	6S

In Figure 4.16, the blue region represents the range of available throttle levels for the given forward speed, while the red line indicates the maximum thrust that the motor and propeller can generate at the given forward speed.

Config 1 provides the advantage of a lightweight motor. However, its smaller propeller diameter results in significantly lower thrust at low speeds compared to the other configurations, making it unsuitable for the mission requirements.

Config 3, featuring a larger diameter and lower-pitch propeller than Config 2, generates higher thrust at low speeds but lower thrust at high speeds. Given these characteristics, configurations for each mission are shown in Table 4.8.

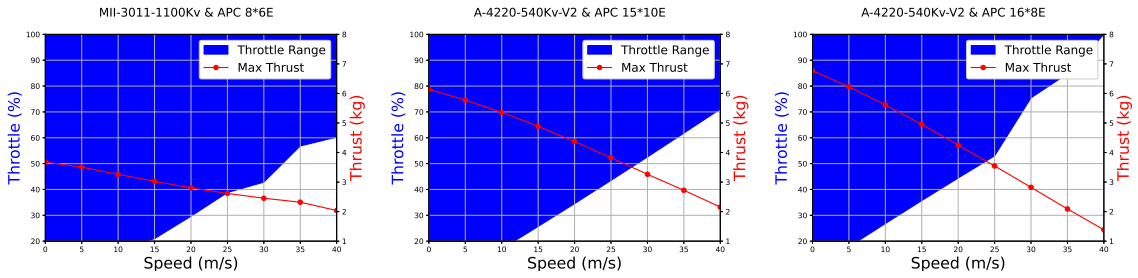


Figure 4.16: **Results of Throttle Range and Max Thrust analysis**

Table 4.8: **Propulsion package of Mission 2, Mission 3**

Mission	Motor	ESC	Battery	Propeller
M2	Scorpion A-4220-540Kv-V2	FLYFUN 130A V5 HV	Vega Graphene 2200mAh 6S 70C LiPo battery	APC 15×10E
M3				APC 16×8E

4.4.3 | Propulsion Analysis

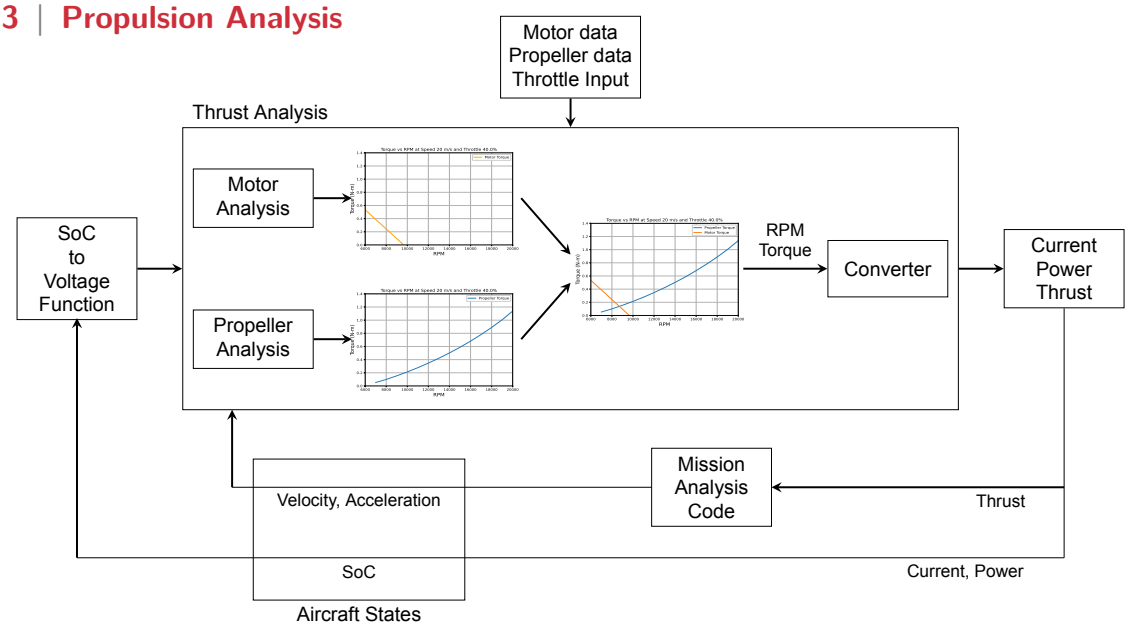


Figure 4.17: **Flow chart of propulsion analysis**

The Propulsion Analysis Module evaluates the performance of an aircraft's propulsion system by calculating motor, propeller, and battery dynamics. The system begins with aircraft states that include velocity, acceleration, and the state of charge (SoC) of the battery.

First, battery voltage is determined based on experimental discharge data. Given a throttle input, the RPM and torque are determined by predefined motor and propeller specifications, as described in Section 4.4.2. Once the RPM and Torque are determined, current, power, and thrust of the system is calculated. Propeller aerodynamic data is used to determine thrust based on the given forward speed and RPM. Additionally, power consumption and current draw are computed.

The power consumption is then used to update the SoC, while thrust is utilized in the mission analysis code to determine velocity and acceleration. The updated aircraft states are subsequently used for the next computation cycle, enabling an iterative analysis of the propulsion system.

Overall, this Propulsion Analysis Module offers a more accurate approach to evaluating aircraft performance by combining electrical and aerodynamic models, as it integrates manufacturer-provided motor, propeller and battery test data into the calculations.

The voltage sag of the battery varies with output current and is not easily calculated. Thus, it was assumed that the voltage would follow a discharge curve with a 30A discharge rate. However, real-world scenarios, such as climbing maneuvers that require higher throttle inputs, will increase the discharge current and thus have a greater voltage sag. These effects are not fully accounted for in the current model. Further refinement and validation through actual flight testing are necessary to confirm the results.

4.5 | Optimization

4.5.1 | Mission Analysis

The total score was calculated making the configurations ”participate” in the competition and letting the maximum objective for each configuration be the maximum objective of the competition. Also, it was assumed that Mission 1 was completed, and that all configurations received a 2.5 bonus score.

Analysis of the results revealed AR as a particularly critical parameter, with a clear optimal value around 4.5 for Mission 2. Score degradation was significant for AR values outside this range. Additionally, the span of the main wing showed a strong correlation with Mission 2 performance, with larger spans generally yielding better results up to the regulatory limit. Other geometric parameters like twist angle and taper ratio showed relatively minor impact on the final scores. Both airfoils under consideration (SG6043 and E216) performed similarly across the analyzed configurations, though the E216 airfoil produced slightly better results in the highest-scoring cases.

Table 4.9: Top 5 Scoring Cases of Mission Analysis

Airfoil	Span (mm)	AR	Taper	Twist	MTOW	Fuel Weight	Flight Time	Objective 2	N_{laps}	Objective 3	Total Score
E216	1800	4.71	0.90	2.0	10.0	5.484	106.1	0.113912	8	12.671506	4.926855
E216	1800	4.71	0.85	2.0	10.0	5.482	106.1	0.113886	8	12.671506	4.926625
E216	1800	4.71	0.90	1.0	10.0	5.484	106.2	0.113802	8	12.671506	4.925886
E216	1800	4.71	0.75	2.0	10.0	5.478	106.1	0.113789	8	12.671506	4.925775
E216	1800	4.71	0.85	1.0	10.0	5.482	106.2	0.113776	8	12.671506	4.925654

4.5.2 | Final Configuration

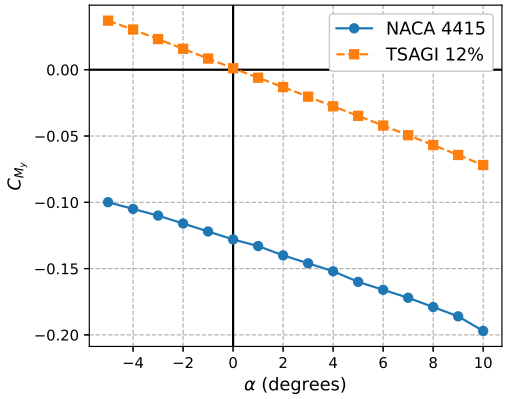
Based on the optimization results, we finalized the aircraft design and mission parameters to achieve the best performance.

Table 4.10: Characteristics of final aircraft design

Aircraft parameters					Mission2 parameters				Mission3 parameters		
Airfoil	Span (m)	AR	Taper	Twist	MTOW (kg)	Max. speed (m/s)	Turn thrust ratio	Level flight thrust ratio	Max. speed (m/s)	Turn thrust ratio	Level flight thrust ratio
E216	1.8	4.71	0.9	2.0	10.0	34	0.7	0.9	24	0.4	0.5

4.6 | Preliminary Design of the X-1 Test Vehicle

4.6.1 | Glider Design



Geometrical design parameters	Values
Airfoil	TSAGI 12%
Wingspan	500mm
Root Chord length	250mm
Tip Chord length	60mm
Taper ratio	0.24
Sweepback angle	43°
Dihedral angle	5°
Elevon position along wingspan	0.2 ∼ 0.9
Elevon width	30mm

Figure 4.18: X-1 Test Vehicle Sizing Results

In the conceptual design phase, the X-1 was determined to have a flying wing configuration. Considering that X-1 is a glider with no thrust, improving stability was set as the top priority in the preliminary design process.

Two factors were considered to ensure longitudinal static stability: $C_{m,0}$ must be positive, and C_{m_α} must be negative. To achieve this, the CG of the X-1 was positioned in front of the aerodynamic center to ensure a negative C_{m_α}. Two configurations were considered to achieve a positive $C_{m,0}$: a negative camber airfoil and a reflexed airfoil. However, the former was rejected due to its poor aerodynamic performance. Instead, TSAGI 12%, a positively cambered airfoil with reflex characteristics, was chosen to enhance longitudinal static stability while maintaining aerodynamic performance.

The trailing edge was designed with a sweepback configuration. Winglets were added to enhance both yaw and roll stability. However, since winglets alone were deemed insufficient for roll stability, a 5° dihedral angle was applied. Additionally, elevons were distributed widely along the span to maximize the controllability of the aircraft.

The preliminary design process was done via OpenVSP. The results were used to evaluate static stability and estimate the center of gravity, ensuring an appropriate static margin. To increase the controllability of the test vehicle, the static margin was set to be lower than 10%.

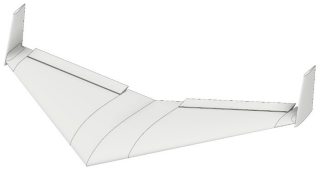


Figure 4.19: X-1 Test Vehicle

4.6.2 | Control and Guidance

X-1 Control System

The X-1 is equipped with a MicoAir743-AIO-35A flight controller with ArduPilot firmware installed. The control surfaces are operated by servos connected to the flight controller. In Auto mode, a predefined setpoint is assigned, and the firmware calculates the target position to follow the loiter path.

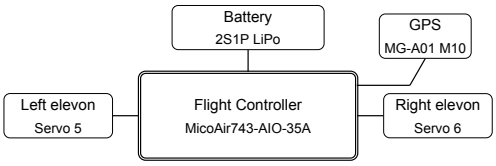


Figure 4.20: Flight control system

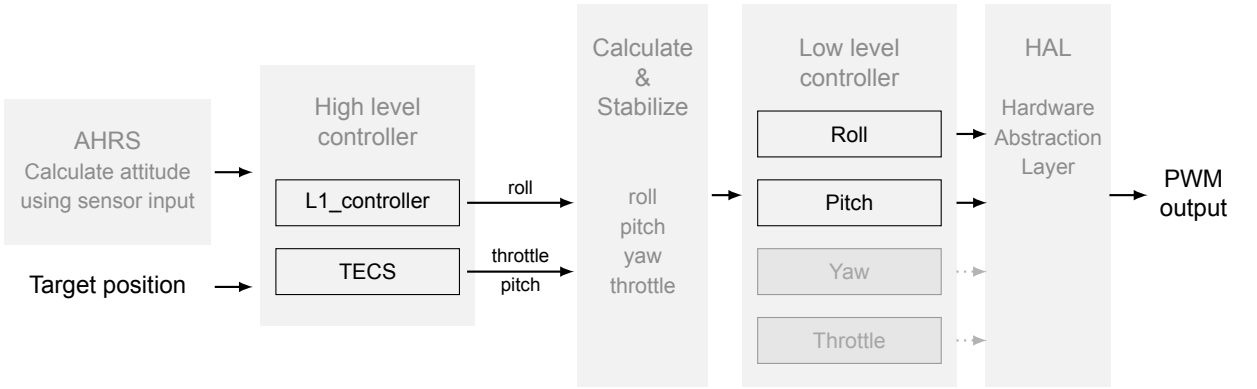


Figure 4.21: **Control system diagram**

When a Loiter command is executed, it is transmitted to the high-level controller. The L1 controller governs lateral navigation by computing the roll angle required to maintain a circular trajectory within the designated radius. Any deviations from the intended path are corrected through continuous roll angle adjustments. For vertical control, the Total Energy Control System (TECS) determines the optimal throttle and pitch settings to maintain the commanded airspeed and altitude. After stabilization adjustments are made, the computed control inputs are relayed to the low-level controller for execution. At this stage, since the X-1 is a motorless glider, the throttle values are always 0.

The target roll and pitch values are converted into servo inputs. Yaw and throttle inputs are disregarded. Then, roll and pitch controls are mixed to control the elevon configuration. Finally, the processed servo inputs are transmitted to the motors through the Hardware Abstraction Layer (HAL).

Guidance of X-1 Flight

The X-1 aircraft was guided to descend while loitering around a designated target point. To satisfy the 180° turn requirement and ensure a stable entry into the bonus zone, the loiter radius was set to less than 33 m. Using the autopilot's soaring system, the glider was controlled exclusively through roll and pitch controllers without any throttle input. The functionality of this system was verified through SITL (Software-in-the-Loop) simulations. Figure 4.22 shows the simulation results, where the test vehicle descends from 100 m to 0 m in a circular trajectory.

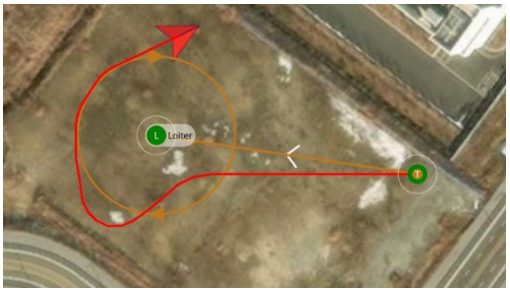


Figure 4.22: **SITL of X-1 plane in QGroundControl**

4.7 | Predicted Performance

4.7.1 | Aerodynamic Analysis

Ansys Fluent was used as the primary tool for aerodynamic analysis. Flow conditions and simulation parameters were established based on predicted level flight conditions as given by Table 4.11. A Kappa-Omega turbulent flow model with the Green-Gauss node method was used alongside polyhedral meshing. A python script was utilized to facilitate the analysis over multiple configurations.

Flow Parameters	Value
Turbulence Model	Kappa-Omega
Reynolds Number	61800
Freestream Velocity	34 m/s
Angle of Attack Range	-3.0° to 12.0°
Temperature	20°C (293.15K)
Pressure	1atm

Simulation Parameters	Value
Surface Mesh Size	0.005m to 0.5m
Number of Boundary Layers	10
Curvature Normal Angle	12
Volume Mesh Size	0.5m max
Volume Mesh Type	Polyhedral
Gradient Scheme	Green-Gauss Node-Based
Convergence Iterations	500

Table 4.11: Flow and Simulation Parameters used in the Ansys Fluid Analysis

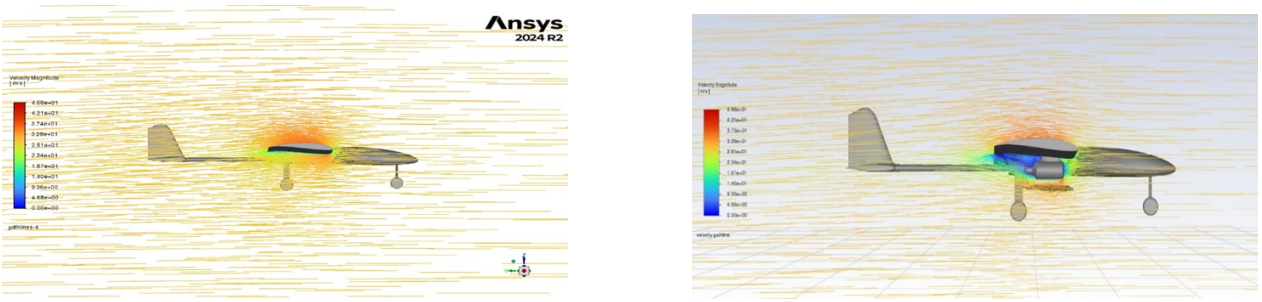


Figure 4.23: Velocity Streamlines around each configuration

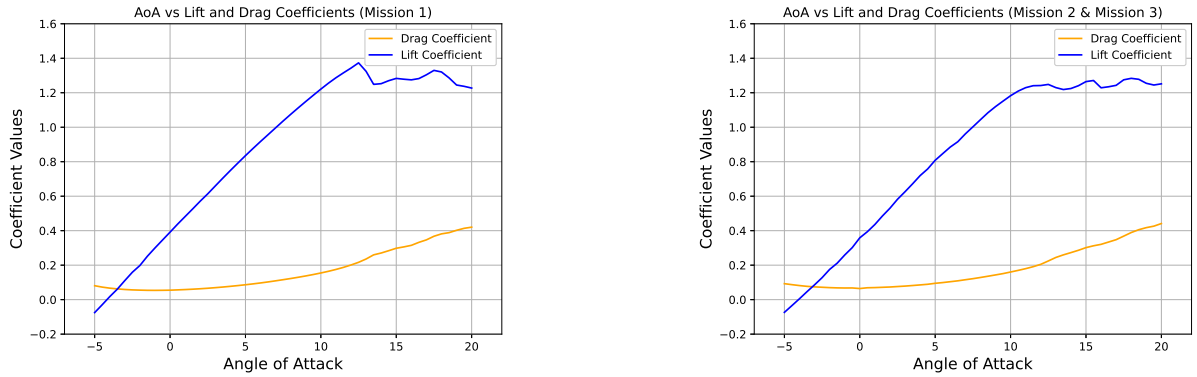


Figure 4.24: Aerodynamic Coefficients (left to right: Mission 1, Mission 3)

Since the fuel tank, X-1, and the launch system affect aerodynamics, an analysis was performed individually for each configuration. As a result, the stall angle was observed to be 12.5° in both configurations. In line with intuition, the Mission 2 and Mission 3 configurations exhibited slightly lower lift and higher drag coefficients compared to Mission 1's configuration. This can also be observed through the flow characteristic (velocity streamlines) around the vehicles.

4.7.2 | Static Stability

Static Stability of the Main Aircraft

Static stability refers to the ability of an aircraft to return to its original flight attitude after being disturbed by external factors, such as wind gusts or changes in control input. This property ensures that the aircraft

Table 4.12: Static stability coefficients of the aircraft analyzed by AVL

	C_{m_α}	C_{m_q}	C_{l_β}	C_{n_β}	C_{n_r}	Static Margin
Requirement Sign	-	-	-	+	-	+
Mission 1	-0.6735	-13.3631	-0.0382	0.0782	-0.1007	0.1508
Mission 2	-0.6752	-13.3714	-0.0500	0.0768	-0.1027	0.1512
Mission 3	-0.6801	-13.3857	-0.0482	0.0770	-0.1026	0.1508

remains in stable flight without excessive pilot intervention, helping to maintain safety during flight. Static stability is a crucial factor for ensuring the aircraft's response to disturbances is predictable and controlled. Table 4.12 contains the static stability coefficients and the static margin of the baseline aircraft, as analyzed using AVL[14].

Static Stability of X-1

Table 4.13: Static stability coefficients of the X-1 Test Vehicle analyzed by AVL

	C_{m_α}	C_{m_q}	C_{l_β}	C_{n_β}	C_{n_r}	Static Margin
Requirement Sign	-	-	-	+	-	+
X-1 Test vehicle	-0.3726	-1.5340	-0.1855	0.0531	-0.0379	0.1013

Since the X-1 has a flying wing configuration, it was deliberately designed with a lower static margin relative to the main aircraft. This configuration, while providing unique aerodynamic benefits, significantly affects the overall stability coefficients, notably resulting in a particularly low $C_{m,q}$. Low $C_{m,q}$ is feasible because, unlike the main aircraft, the X-1 operates without a propulsion system and thus experiences gentler acceleration during flight. As a result, the X-1 can still maintain stability and control.

4.7.3 | Dynamic Stability

Dynamic Stability of the Main Aircraft

To examine the dynamic stability of the aircraft, a linearized aerodynamic model was generated in AVL. The eigenvalues of the aircraft's equations of motion were then computed for each mission. The following figures show the distribution of these eigenvalues in the complex plane.

From Figure 4.25, it can be observed that Roll(X), Short Period(O), Dutch Roll(□), Phugoid(◊), and Spiral(+) modes are clearly distinguished for each case. The table below presents the characteristic values for Mission 2 in level flight (case 1), banking (case 2), and climbing (case 3) conditions. Since the values for Mission 1 and Mission 3 are similar, only Mission 2, which represents the most extreme case, is shown.

Table 4.14: Characteristic Value Pairs for Mission 2

	Roll	Short Period	Dutch Roll	Phugoid	Spiral
Cruise	$\omega_n = 15.3174$ $\zeta = 1$	$\omega_n = 13.3458$ $\zeta = 0.6935$	$\omega_n = 6.5714$ $\zeta = 0.1115$	$\omega_n = 0.3177$ $\zeta = -0.0157$	$\omega_n = 0.0388$ $\zeta = -1$
Banking	$\omega_n = 12.3337$ $\zeta = 1$	$\omega_n = 12.2813$ $\zeta = 0.6573$	$\omega_n = 7.5803$ $\zeta = 0.1497$	$\omega_n = 1.2947$ $\zeta = -0.0063$	$\omega_n = 0.2107$ $\zeta = 1$
Climbing	$\omega_n = 6.9093$ $\zeta = 1$	$\omega_n = 7.0062$ $\zeta = 0.6392$	$\omega_n = 5.1417$ $\zeta = 0.1217$	$\omega_n = 0.8512$ $\zeta = 0.1219$	$\omega_n = 0.0313$ $\zeta = 1$

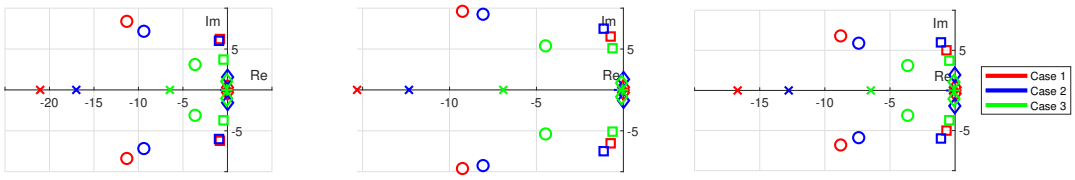


Figure 4.25: **Eigenvalues of Main Aircraft (left to right: Mission 1, Mission 2, Mission 3)**

From the eigenvalue plot in Figure 4.25, most eigenvalues are located in the left-half plane, indicating stable or damped modes. However, in certain modes, such as the Spiral mode, eigenvalues appear in the right-half plane, indicating the presence of unstable dynamic mode in the linearized model. Although there are unstable modes, the phugoid and spiral modes have long time constants. As a result, they can be sufficiently stabilized through pilot control.

Dynamic Stability of X-1

Table 4.15: **Characteristic Value Pairs for the X-1**

Mission2 case	Roll	Short Period	Dutch Roll	Phugoid	Spiral
Cruise	$\omega_n = 49.9914$ $\zeta = 1$	$\omega_n = 16.4075$ $\zeta = 0.4413$	$\omega_n = 14.8364$ $\zeta = 0.1691$	$\omega_n = 1.1279$ $\zeta = -0.0509$	$\omega_n = 0.0906$ $\zeta = 1$

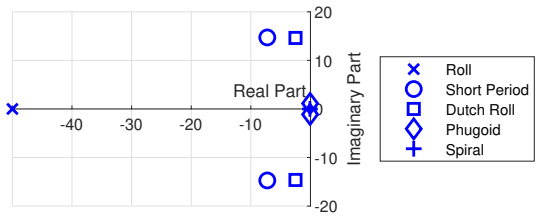


Figure 4.26: **Eigenvalues of X-1**

Unlike the aircraft, the X-1's eigenvalues are all in the left-half plane, indicating stable flight in all modes.

4.7.4 | Predicted Mission

The predictions for flight time and mission score are presented in Table 4.16.

Figure 4.27 illustrates the propulsion system analysis for Mission 3, where powertrain performance is the most critical factor. The voltage plot demonstrates that the battery voltage reaches its lower limit at the end of the flight, which represents the optimal optimization point in terms of both time and energy utilization. Additionally, throttle level, thrust, SoC, and current were plotted to provide pilots with a control manual while ensuring safe flight by verifying that the propulsion system does not exceed its operational limits.

Table 4.16: Predictions for flight time and mission score

	Mission 1	Mission 2	Mission 3
Takeoff weight	4.52	10	4.72
Flight Time	85.1	106.1	263.9
Number of Laps	3	3	8
Predicted Score	1	2	2.93

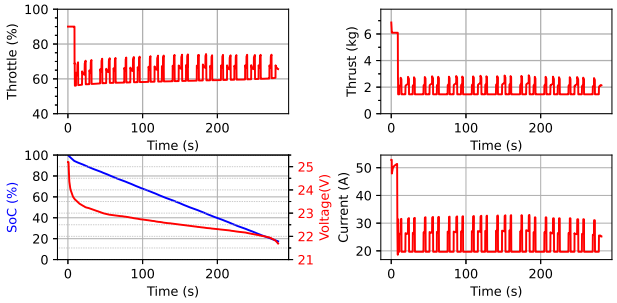


Figure 4.27: Propulsion system analysis

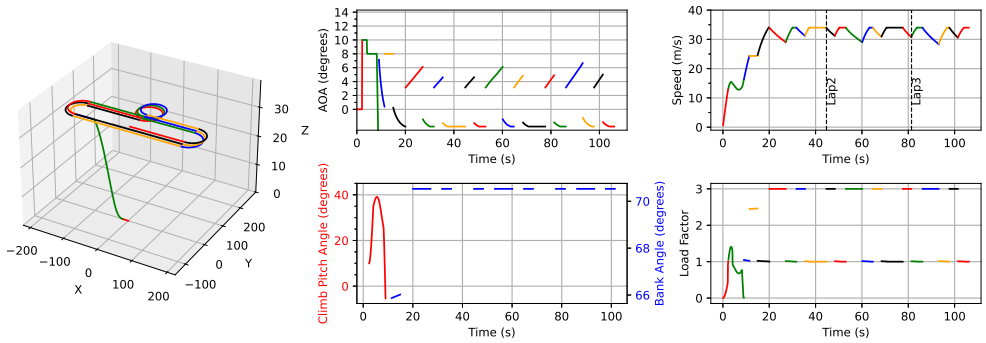


Figure 4.28: Analysis of Mission 2 (Total flight time: 106.1 s, N_laps: 3)

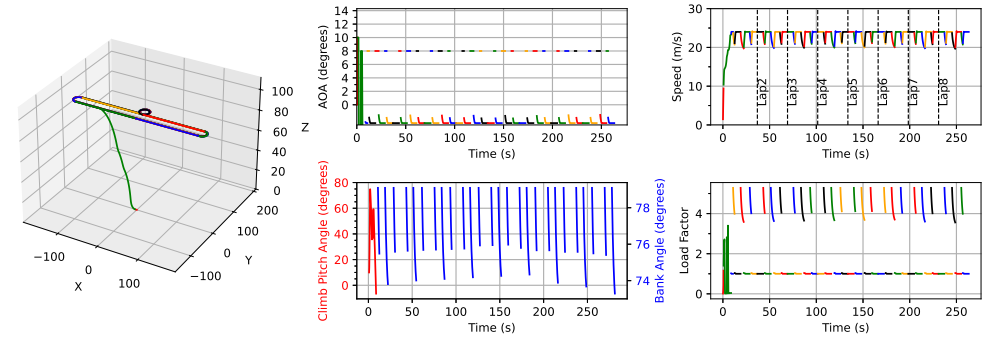


Figure 4.29: Analysis of Mission 3 (Total flight time: 263.9 s, N_laps: 8)

5 | Detail Design

Table 5.1: Aircraft and X-1 Manufacturing Details

Aircraft			
Wing		Fuselage / Tail boom	
Airfoil	E216	Total Length	5.53 ft (1.686m)
Span	5.9 ft (1.8m)	Max Width	11.4 in (0.290m)
MAC	1.24 ft (0.378m)	Max Height	5.83 in (0.148m)
Root Chord	1.31 ft (0.399m)	Tail boom Length	3.00 ft (0.914m)
Tip chord	1.18 ft (0.359m)	Tail boom Width	0.945 in (0.024m)
AR	4.71	Controls	
Area	7.34 ft² (0.682m²)	Receiver	Cooltech RFA16 Futaba Fasst Receiver
Dihedral Angle	5°	Servo	Hitec D145SW Servo
Horizontal Tail		Vertical Tail	
Airfoil	NACA 0008	Airfoil	NACA 0009
Span	16.22 in (0.412m)	Span	7.24 in (0.184m)
Chord - base	8.11 in (0.206m)	Chord - base	8.11 in (0.206m)
Chord - tip	8.11 in (0.206m)	Chord - tip	4.88 in (0.124m)
AR	4	AR	1.12
Area	263.5 in² (0.170m²)	Area	46.5 in² (0.030m²)
Mission No.	Battery	Motor	Propeller
1	Vega Graphene 2200mAh 6S 70C LiPo battery	Scorpion A-4220-540Kv-V2	APC 15×10E
2			
3			APC 16×8E

X-1	
Wing	
Airfoil	TSAGI 12%
Span	15.75 in (0.40m)
MAC	6.69 in (0.17m)
Root Chord	9.84 in (0.25m)
Tip chord	2.36 in (0.06m)
AR	3.2
Area	120 in² (0.0775m²)
Dihedral Angle	5°
Sweep Angle	43°
Twisted Angle	0°
Winglet Span	2.76 in (0.07m)
Winglet Angle	90°
Electronics	
Flight Controller	Micoair H743 + 35A ESC AIO
GPS	MicoAir MG-A01 M10 GPS
Servo	EMAX ES09MA Servo
Battery	LiPo 450mAh 2S JST RC Battery K969
LED	3W SMD LED

5.1 | Main Aircraft Detail Design

The aircraft is designed to meet mission requirements while facilitating manufacturing processes. The skin was molded using vacuum bagging method, for high strength-to-weight ratio as well as its ability to form complex curved shape. Internal reinforcements were printed out of PA-CF, for its high impact resistance. The main wing and fuselage section form a blended wing-body shape to reduce drag , and allow the wing and fuselage to be molded into a single part. The aircraft is designed to endure up to 3g accelerations under maximum payload conditions (i.e. Mission 2 conditions), based on FORCAST analysis.

5.1.1 | Fuselage

The fuselage design is based on the GOE601 airfoil. Thus, the fuselage can generate lift and reduce wing loading. The airfoil was chosen due to its maximum camber being located in the middle of the airfoil, and its high thickness which is needed for the fuselage. The fuselage cross section is rounded and gradually blended to the wing section to minimize drag. Cooling ducts are installed on the fuselage to manage the temperature of the electronics. The canopy is cut out to ensure easy maintenance, and can be installed on the fuselage using multiple sliding locking pins. Bulkheads and formers are inserted to create a monocoque while serving as a frame for internal components.



Figure 5.1: Fuselage Airflow

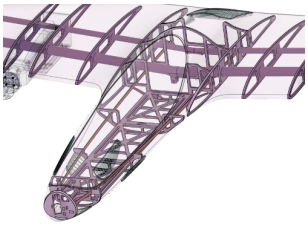


Figure 5.2: Fuselage frame

5.1.2 | Wing

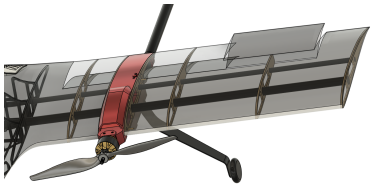


Figure 5.3: **Wing Structure and Control Surfaces**

The wing features a taper ratio of 0.9 and 5° of dihedral angle. The root of the wing is connected to the fuselage, while the outer wing sections are detachable.

A section of the wing acts as the motor, landing gear, boom, and pylon mount simultaneously, designed to impart the force to the wing spar. To prevent wing deformation from bending moments caused by lift, load is transferred from skin to the ribs to two carbon pipes placed to serve as spars.

The main wing features two control surfaces: ailerons and flaps. The flaps had to be split spanwise due to the torque limitations of the servos. The split flap mechanism was used for the flaps for easier manufacturing. Ailerons take the shape of frise aileron, and are programmed to have different deflection angle to reduce adverse yaw. The servos were initially installed on the hardpoints between ribs, which were removed in favor of directly attaching to the skin via mounts for weight reduction and manufacturing simplicity.

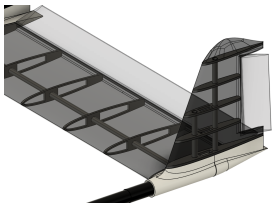


Figure 5.4: **Empennage Structure and Control Surfaces**

5.1.3 | Empennage

The empennage is constructed using the same structure and materials as the main wing. The elevators and rudder are manufactured in a similar fashion to the ailerons, for manufacturing simplicity and easier CG calculation.

5.1.4 | Landing Gear

A commercial nose landing gear with steering capability and spring suspension was selected to facilitate taxiing on the runway and absorb landing impact respectively. The height of all the landing gears are set at 200 mm to ensure easy access to the payload and the glider, as well as making sure the tail doesn't strike the runway or cause the aircraft to nose up during takeoff and landing. The nose landing gear is mounted on a hard point located between the first and second bulkheads. The rear landing gears are attached on the motor mount and are angled outward by 35°, which absorbs landing impact forces with the material's flexibility. The nose and rear landing gears were spaced with 6:1 distance ratio from CG point, ensuring easy takeoff and taxiing.



Figure 5.5: **Landing Gears**

5.1.5 | Payloads

3 main factors were considered for the external fuel tanks and the pylons: weight reduction, easy installation/removal, and secure mounting. A structure inspired by a dovetail rail was used to meet 3 requirements, where the user can quickly slide either the pylon or the cover part from the rear, and secures it to the mount using a set of screws. The weight is saved by adding the dovetail features to preexisting part, and secure mounting is guaranteed by using a long rail and mount.

5.1.6 | Assembly

The main aircraft can be divided into 8 parts - fuselage, left and right wing, twin boom, horizontal tail, and vertical tails for easy transport and assembly.



Figure 5.6: **Assembly**

5.2 | X-1 Test Vehicle Detail Design

The X-1 test vehicle was designed primarily with a focus on weight reduction. To achieve this, a flying-wing configuration was chosen, and only two control surfaces are used to control pitch and roll. the release mechanism was engineered to function without additional parts attached to the X-1. An optical endstop switch is added on top to detect when it is released. To facilitate maintenance, the X-1 hatch was designed to be removable.

5.2.1 | Release Mechanism

Three key design factors for the release mechanism are as follows: 1) Simple attachment for the Ground Mission 2) The mechanism should integrate into X-1 without adding extra weight to X-1, and 3) Reliable and Fast operation.

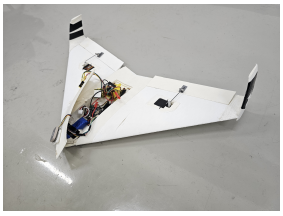


Figure 5.7: **Fully 3D printed X-1 prototype**



Figure 5.8: **Improved structural design of the X-1**

The release mechanism can be easily mounted below the fuselage via 4 screws. Since the X-1 is very lightweight and has lower cruise speed compared to the aircraft, it could show unexpected maneuvers and collide with the aircraft if released at the same AoA. To prevent this, the release mechanism was designed to ensure that the X-1's AoA would be lower than that of the aircraft during release. However, keeping the lowered AoA during flight would increase the total drag, which meant that the X-1 should be kept parallel to the aircraft before release. Thus the release mechanism was designed to keep the X-1 parallel to the aircraft during normal flight, but tilt the X-1's nose downward during release, allowing

the X-1 to naturally separate from the aircraft by the aerodynamic forces acting at the point of release. A part of the release mechanism obscures the optical endstop switch located on top of X-1. Release is detected by the optical endstop switch being no longer obscured.

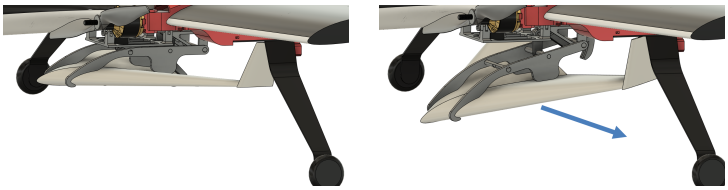


Figure 5.9: Release Mechanism

5.3 | Structural Analysis

The Ansys Fluent solution for maximum lift is revisited for a fluid-solid interaction (FSI) analysis. The aircraft stalls at 12.5° AoA, as detailed in Section 4.7.1. To simulate extreme conditions, flight at this stall angle was assumed at 35 m/s. FORCAST analysis indicates that the highest load occurs during the turning maneuver, with a maximum speed of 34 m/s and an angle of attack of 8°. To account for extreme circumstances, the structural analysis was conducted up to 10g load conditions. The maximum stress was confirmed to be well below the ultimate tensile stress of materials used, confirming that the structure can withstand extreme forces beyond mission expectations.

5.3.1 | Visualized Stress Distributions on Wing, Empennage and Connection Rod

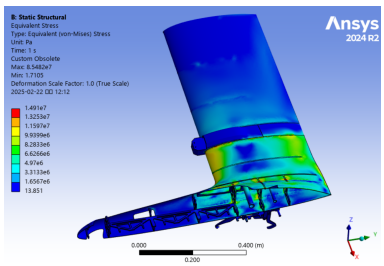


Figure 5.10: Main Wing and Fuselage von-Mises Stress

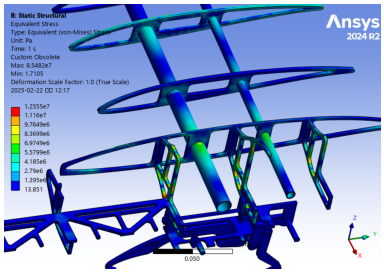


Figure 5.11: Main Wing and Fuselage Inner structure von-Mises Stress

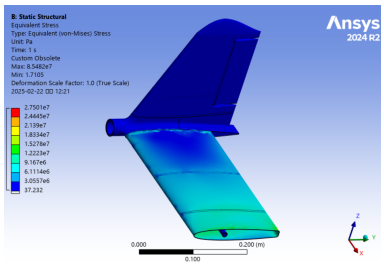


Figure 5.12: Empennage von-Mises Stress

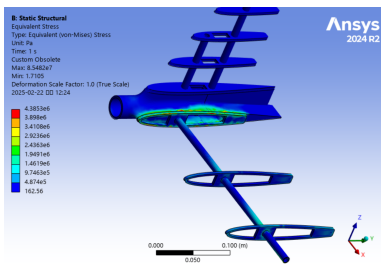


Figure 5.13: Empennage Inner Structure von-Mises Stress

5.3.2 | Stress and Failure Condition Evaluation

The maximum von-Mises stress across the entire structure is achieved on the center fuselage-wing transition piece at 85.48 MPa as shown in Figure 5.10. The maximum stress experienced by the inner support structure is rated at 14.91 MPa as shown in Figure 5.11. Major stresses due to lift force production are mainly distributed around the wing root. Local maximum von-Mises stress were also observed on other sections. The local maximum stress on the empennage was 275 MPa on the outer frame of the tail wing, with stresses mainly distributed in the piece that connects the horizontal stabilizers to the vertical stabilizers. The maximum von-Mises stress on the connecting carbon rod was 8.90 MPa, primarily concentrated at junction points where carbon rods of different diameters connect. All analyses were conducted under 10g conditions. Assuming a linear elastic response without buckling, loads were applied by mapping the Ansys Fluent pressure distribution solutions on the structure to simulate an extreme mission scenario.

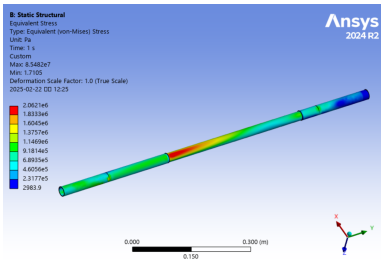


Figure 5.14: **Carbon Pipe von-Mises Stress**

Table 5.2: **Stress Analysis Results at 10g**

Material	Used in	Tensile Strength[15] (MPa)	Maximum stress achieved in part (MPa)	Factor of Safety
PA-CF	Inner Support Pieces	140	14.91	9.39
Carbon Fiber & Fiber Glass Composite	Fuselage and Main Wing Surface	440	85.48	5.15
3K Carbon Fiber	Carbon Rods	3650	8.90	410.11

As shown in table 5.2, all components can withstand the given flight conditions, with the lowest factor of safety being 5.15. However, real testing is needed to account for stress concentrations caused by local structural irregularities arising from manufacturing imperfections and fiber alignment.

5.4 | Weight and Balance

The CG is located at 96.3 mm from the main wing's leading edge. A high static margin of 20% was decided as a high-thrust motor was selected to achieve a high thrust-weight ratio. For stable landing, the landing gear was positioned so that the CG point would be located at a 6:1 ratio between the nose and the rear landing gears. Since the payload affects the CG position, the battery placement was designed to be adjustable to maintain the desired CG point. The battery's center of gravity can be moved from 230 mm to 400 mm with the internal fuel tank attached, and 0 mm to 400 mm without it.

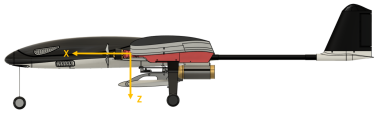


Figure 5.15: **Aircraft Coordinate System**

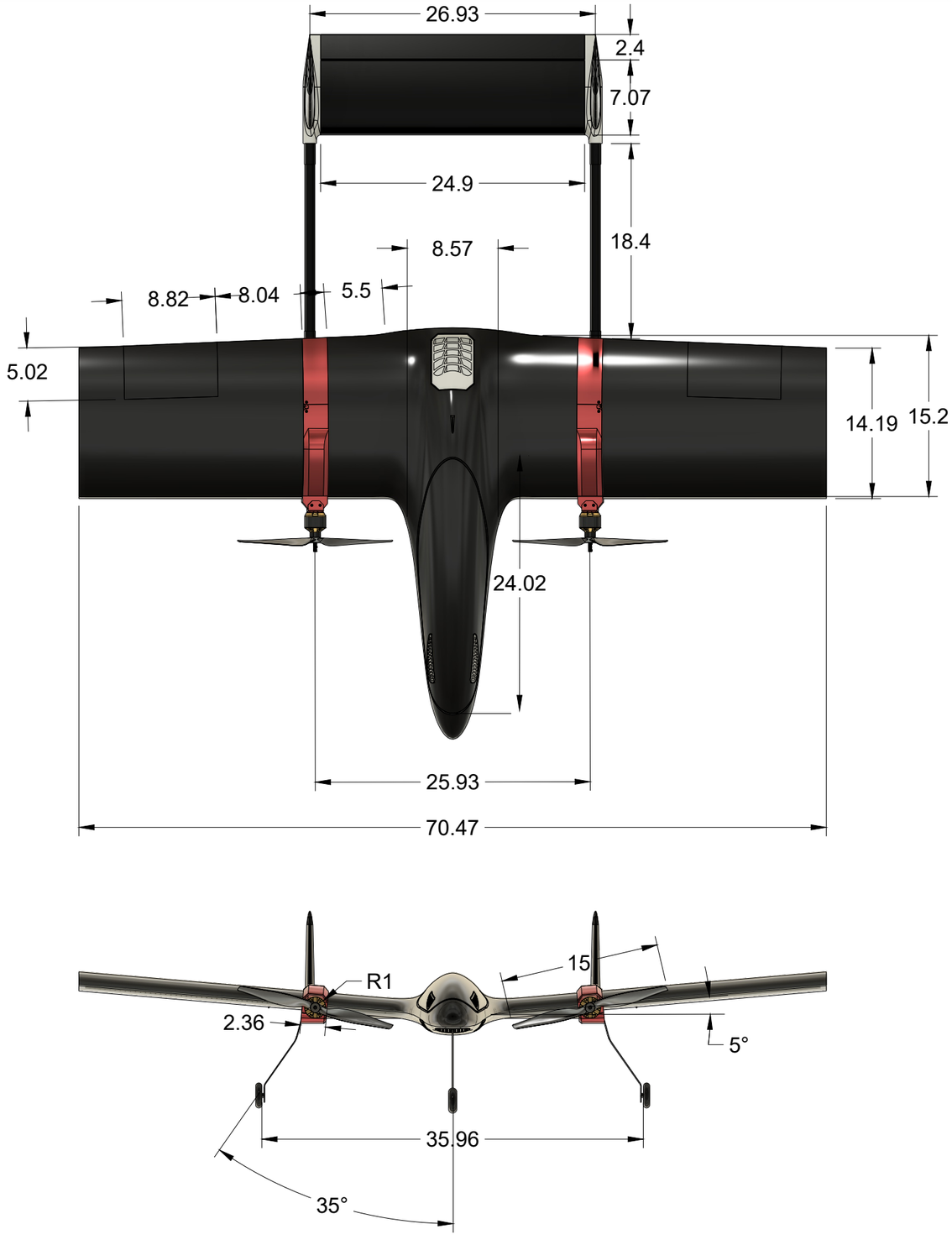
Table 5.3: Weights and Balances

Mission	Aircraft Component	Mass		X Displacement		Z Displacement	
		[lb]	[g]	[in]	[mm]	[in]	[mm]
General	Fuselage and Boom	2.072	1036.0	-14.1	-358.147	-0.96	-24.290
	Wing	0.564	282.0	-7.21	-183.116	-2.78	-70.570
	Horizontal Tail	0.388	194.0	-37.85	-961.397	-0.24	-5.998
	Vertical Tail	0.096	48.0	-39.23	-996.438	-5.03	-127.872
	Nose Landing Gear	0.070	35.0	19.96	507.000	6.94	176.270
	Body Landing Gear	0.100	50.0	-7.66	-194.500	5.26	133.480
	Receiver	0.034	17.0	-5.18	-131.600	-0.58	-14.610
	Motor	0.576	288.0	2.46	62.460	-1.16	-29.486
	ESC	0.443	221.5	11.25	285.720	-0.64	-16.150
Mission 1	Battery	0.686	343.0	15.71	399.000	-1.02	-26.000
	Propeller	0.096	47.9	3.30	83.700	-1.16	-29.486
	Total	7.414	3706.8	-3.79	-96.300	0.00	0.000
Mission 2	Battery	0.686	343.0	15.71	399.000	-1.02	-26.000
	Propeller	0.096	47.9	3.30	83.700	-1.16	-29.486
	Internal Fuel Tank	5.200	2600.0	3.93	99.910	-1.35	-34.340
	External Fuel Tank	2.600	1300.0	-3.93	-99.910	1.05	26.620
	Release Mechanism	0.140	70.0	-4.98	-126.605	2.39	60.780
	X-1 Test Vehicle	0.460	230.0	-3.79	-96.300	3.54	90.000
	Total	19.977	9988.6	-3.79	-96.300	0.00	0.000
Mission 3	Battery	0.686	343.0	15.71	399.000	-1.02	-26.000
	Propeller	0.104	51.9	3.30	83.700	-1.16	-29.486
	Internal Fuel Tank	0.080	40.0	3.93	99.910	-1.35	-34.340
	External Fuel Tank	2.600	1300.0	-3.93	-99.910	1.05	26.620
	Release Mechanism	0.140	70.0	-4.98	-126.605	2.39	60.780
	X-1 Test Vehicle	0.460	230.0	-3.79	-96.300	3.54	90.000
	Total	14.873	7436.6	-3.79	-96.300	0.00	0.000

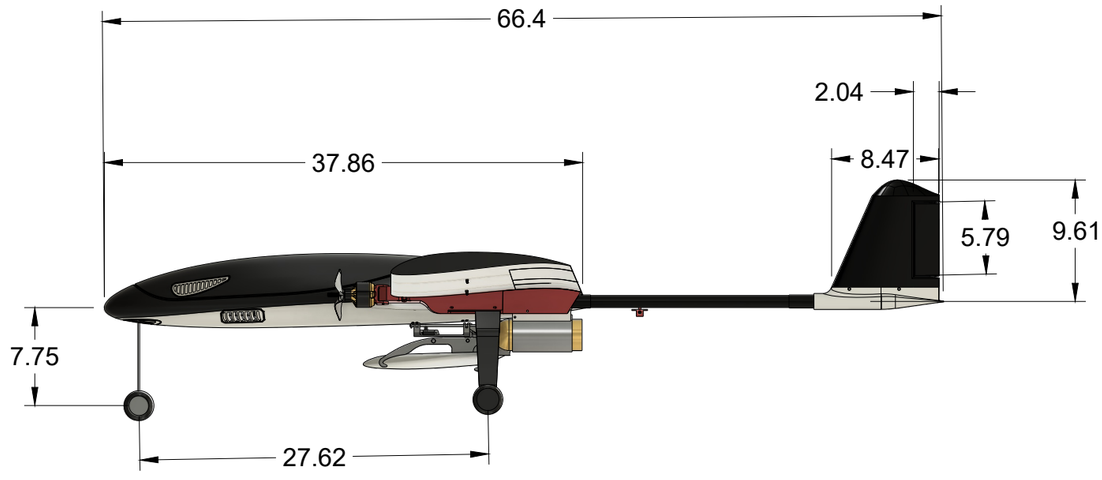
5.5 | Flight Performance & Mission Performance of Final Aircraft

Table 5.4: Flight performance data

	M1	M2	M3	GM
$C_{L_{max}}$	1.33	1.25	1.25	-
$(L/D)_{max}$	9.71	8.55	8.55	-
Stall Angle	12.5	12.5	12.5	-
L/D_{max} Angle	4.5	5.5	5.5	-
MTOW (kg)	4.145	10	4.577	-
Wing Loading	6.08	14.66	6.71	-
Carried Payload	-	Fuel Tanks, Pylon, X-1, Release Mechanism	Fuel Tanks (Empty), Pylon, X-1, Release Mechanism	-
V_{stall} (m/s)	9.47	16.69	11.29	-
$V_{level\ flight}$ (m/s)	30.05	34.52	23.56	-
Number of Laps	3	3	8	-
Mission Time	84.8	110.6	271.5	30







Seoul National University; Gyeongsang National University; Konkuk University

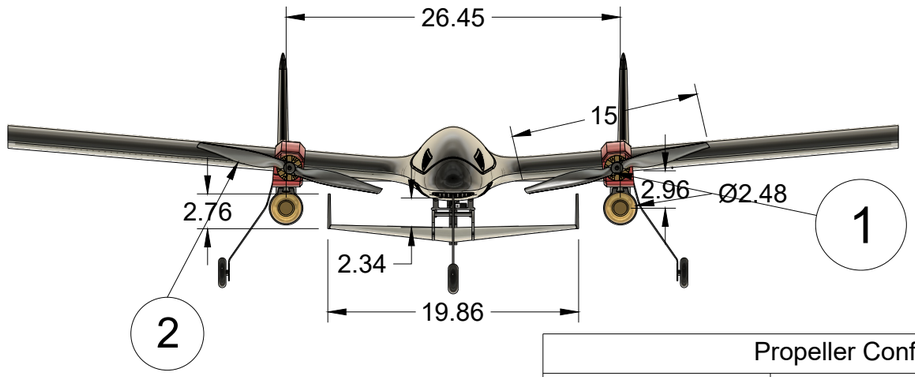
Team Korea

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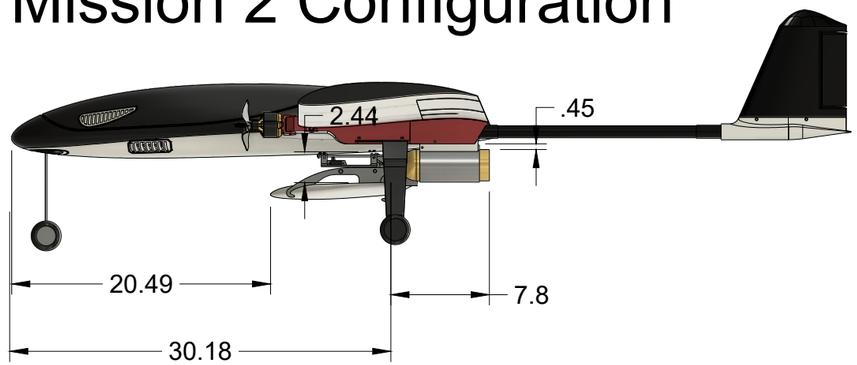
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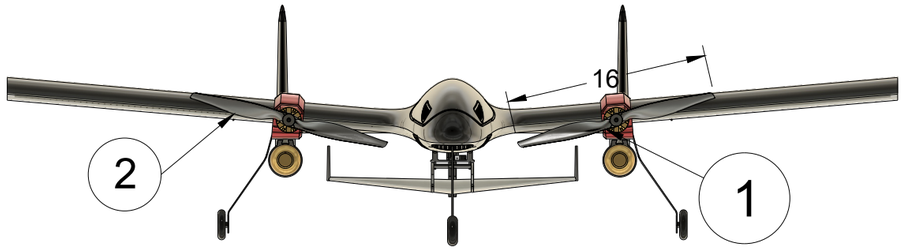
SCALE 1:12	Unit Inch	SHEET 1/2

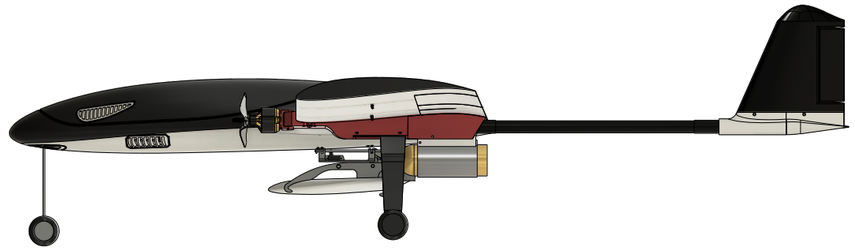


Propeller Configuration		
Part No.	Name	Qty
1	APC 15x10E	1
2	APC 15x10EP	1

Mission 2 Configuration







Propeller Configuration		
Part No.	Name	Qty
1	APC 16x8E	1
2	APC 16x8EP	1



Mission 3 Configuration

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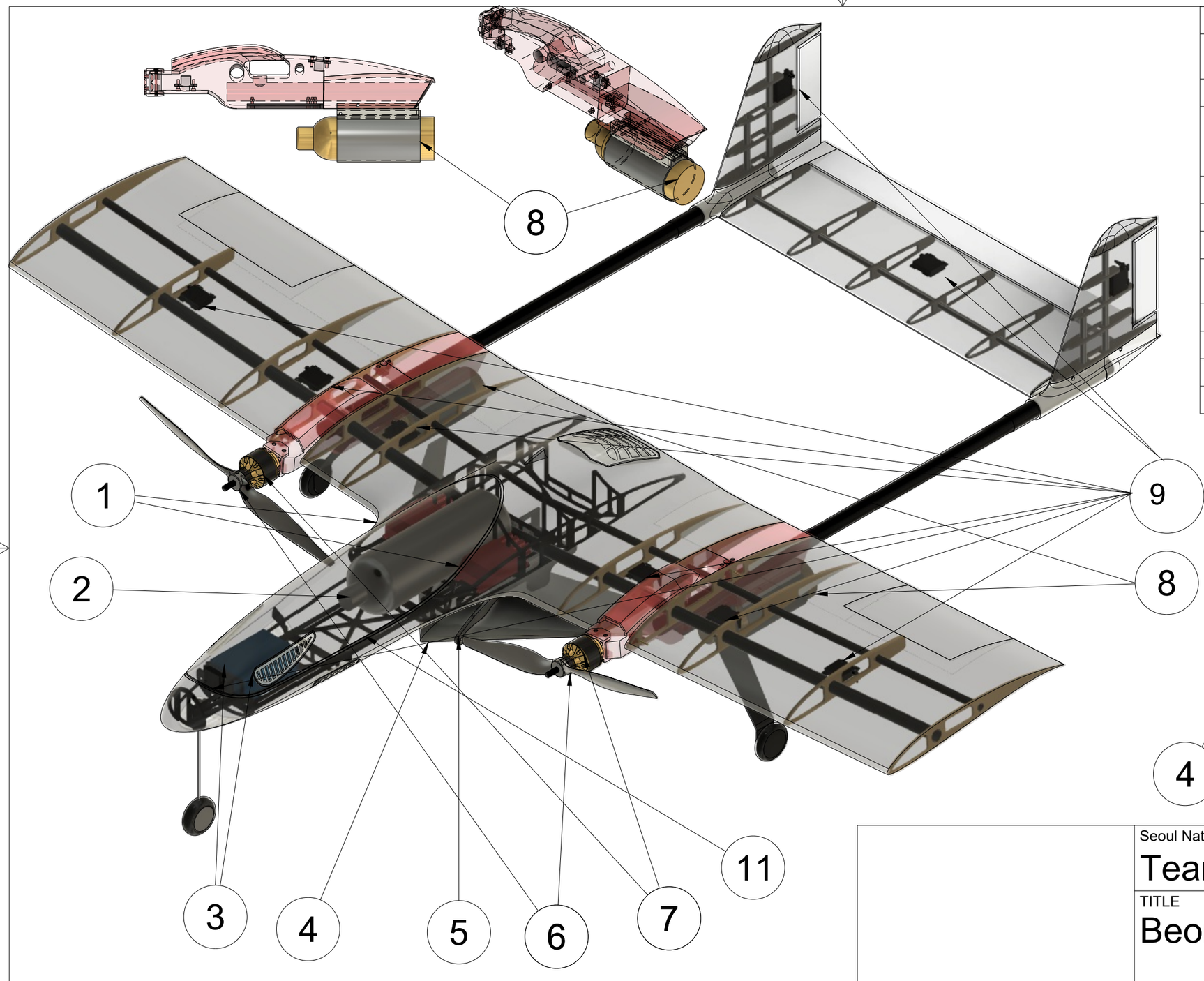
Team Korea

TITLE

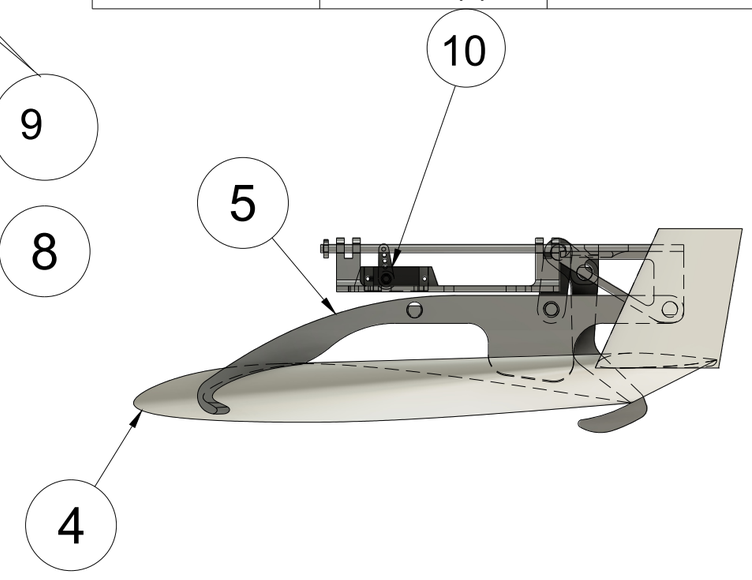
BeomGoRae

SIZE	Mission	DWG NO	REV
B	2&3	Systems Overview	3

SCALE 1:12 | Unit Inch | SHEET 2/2



Part No.	Name	Qty
1	FLYFUN 130A HV OPTO V5 ESC	2
2	1.5L Bottle	1
3	Vega Graphene 2200mAh 6S 70C LiPo battery	2
4	X-1	1
5	Release	1
6	APC 16×8E	2
7	Scorpion A4220 540KV V2	2
8	0.7L Bottle	2
9	Hitec HS-D145SW	10
10	EMAX ES09MA	1
11	Velcro Strap plate	1



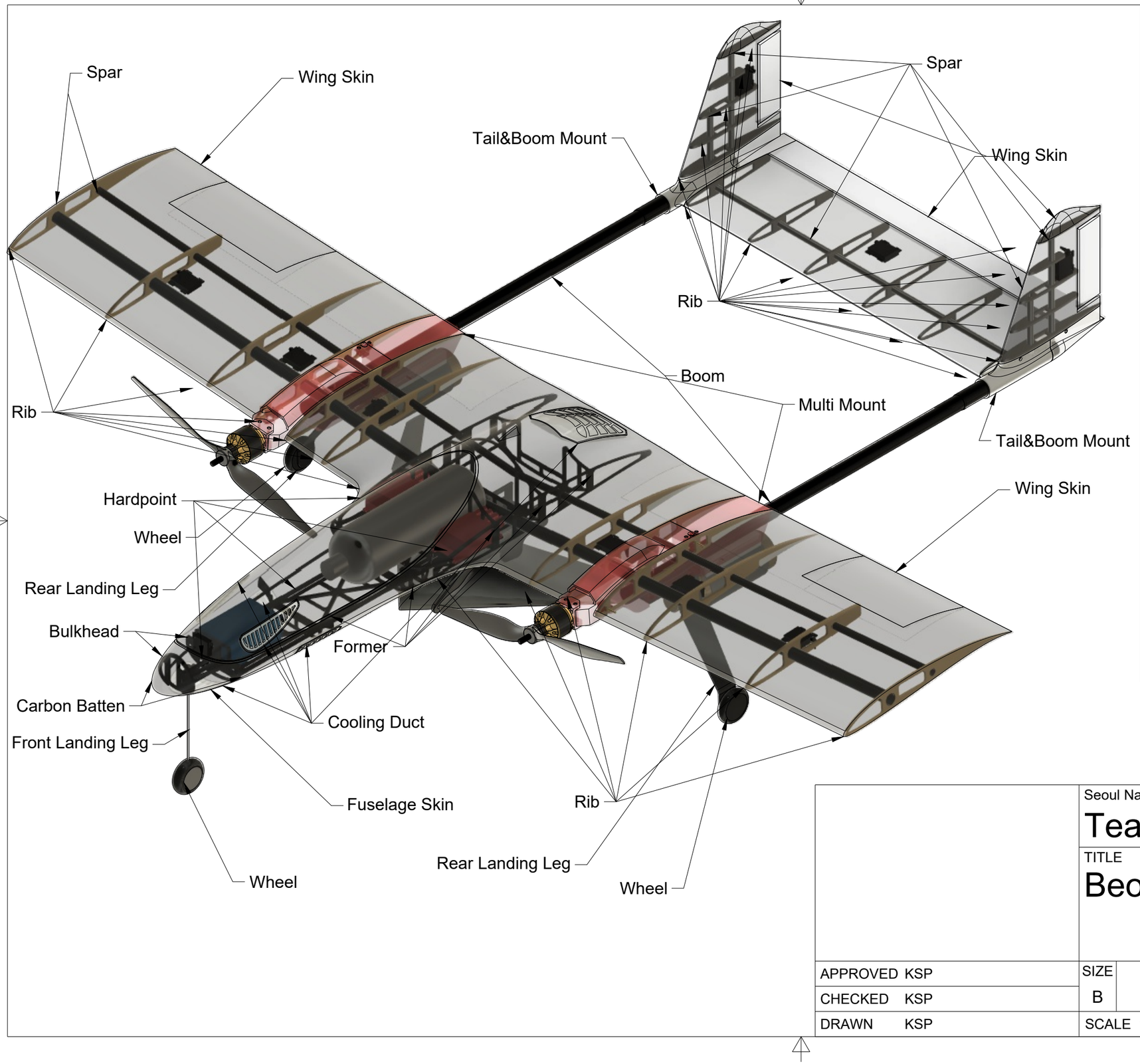
Seoul National University & Gyeongsang National University & Konkuk University

Team Korea

TITLE

BeomGoRae

APPROVED KSP		SIZE	CODE	DWG NO	REV
CHECKED KSP		B		Systems Arrangement & Payload Accommodation	3
DRAWN KSP	21/02/2025	SCALE 1:6	WEIGHT	SHEET 1/1	



Name	Material	Qty
Fuselage Skin	Carbon Fiber Reinforced Plastic	1
Wing Skin	Carbon/Glass fiber Composite	5
Bulkhead/Former	PA-CF	7
Rib	PA-CF	38
Multi Mount	PA-CF	2
Tail&Boom Mount	PA-CF	2
Cooling Duct	PLA	5
Wheel	Sponge	3
Rear Landing Leg	Crbon Fiber Reinforced Plastic	2
Front Landing Leg	Steel	1
Hardpoint	0.7L Bottle	4
Boom	Carbon Fiber Reinforced Plastic	2
Spar	Carbon Fiber Reinforced Plastic	5
Carbon Batten	Carbon Batten	2

Seoul National University & Gyeongsang National University & Konkuk University

Team Korea

TITLE

BeomGoRae

APPROVED KSP	SIZE	CODE	DWG NO	REV
CHECKED KSP	B		Structural Arrangement	3
DRAWN KSP	SCALE 1:6	WEIGHT	SHEET 1/1	

6 | Manufacturing

6.1 | Manufacturing Milestone

Table 6.1: Manufacturing Milestone

AIAA DBF 2024-2025	November				December					January				February				March					April			
	10	17	24	06	01	08	15	22	29	05	12	19	26	02	09	16	23	02	09	16	23	30	06	13	20	27
Prototype 1																						Planned Task				
3D Printing																						Planned Sub-Task				
Bonding by Parts																						Planned Sub-Task				
Final Assembly																						Key Dates				★
Prototype 2																										
Wing Mold Printing and Sanding																										
Wing Lamination																										
Fuselage Mold Printing and Sanding																										
Fuselage Lamination																										
Overall Finishing																										
Final Assembly																										
Prototype 3																										
Wing Mold Printing and Sanding																										
Wing Lamination																										
Fuselage Mold Printing and Sanding																										
Fuselage Lamination																										
Landing Gear Mold Printing and Sanding																										
Landing Gear Lamination																										
Overall Finishing																										
Final Assembly																										
Competition Aircraft																										
X-1 Prototype																										
X-1 Final Model																										
AIAA DBF Competition																								★		

6.2 | Manufacturing Process Investigated

To develop an aircraft optimized for mission requirements, various manufacturing methods including 3D printing, composite material construction, and plywood & balsa construction were considered. The selection process between the methods was based on factors such as manufacturability, strength, and weight. A detailed comparison of the employed methods is presented below.

6.2.1 | 3D Printing

Using Fused Deposition Modeling (FDM) 3D printing simplifies aircraft manufacturing and reduces weight with the suitable filament selection, while facilitating maintenance. The process consists of four steps: CAD design → slicing → printing → assembly, making it highly efficient.

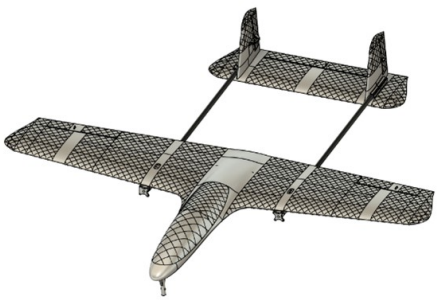


Figure 6.1: Aircraft Designed with Consideration for Vase Mode

Initially, we planned to print the entire aircraft using LW-PLA, a lightweight filament with a density of 0.54 g/cm^3, approximately half the density of standard PLA filaments. Multiple attempts were made to overcome its limitations such as weak strength and poor print quality, by manually designing the infill pattern and covering the skin with a layer of glass microfiber. However, even those improvements proved the material insufficient to be used for the aircraft.

6.2.2 | Composite Material Manufacturing

For Mission 2, where the aircraft must fly with the payload attached, structural rigidity is crucial due to high flight loads. Composite lay-up provides superior strength without significant weight gain. By laminating directly onto 3D-printed molds, complex curved geometries can be accurately fabricated. Carbon fiber and glass fiber are laminated onto molds using vacuum bagging to form the aircraft's skin. Although it takes more time and is costly, composites maximize payload capacity, making them ideal for this mission.

6.2.3 | Balsa and Plywood Construction

The wood construction method, using balsa and plywood, has been widely used for lightweight RC aircraft due to its high strength shown in low density. Ribs and bulkheads are laser-cut, covered with 1mm balsa skin, and filmed with Monokote. This method offers ease of fabrication, simple maintenance, and cost-effective lightweight construction. However, this method requires a laser cutter. Also, the material is fragile and susceptible to moisture exposure so must be handled with care.

6.2.4 | Selection Process

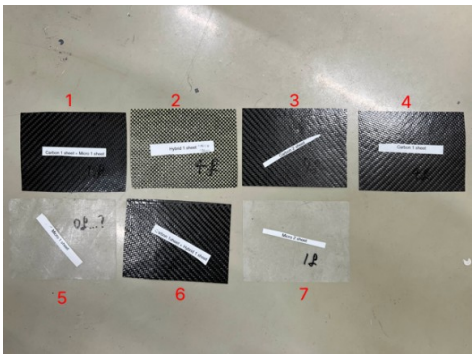
Based on considered manufacturing methods, a trade study was conducted to determine the most suitable manufacturing approach and materials for the aircraft. Various factors were evaluated according to the mission requirements using an AHP with a pairwise comparison matrix. Each factor was assigned a score between 1 and 5 to establish priorities and weightings. The selection criteria for the manufacturing method were as follows:

1. Manufacturability: Better manufacturability allows rapid and straightforward production of multiple prototype and backup aircraft.
2. Strength: Materials with higher strength-to-weight ratio can be used to further reduce weight.
3. Weight: Aircraft weight directly affects flight performance and battery efficiency. Minimizing weight was prioritized to maximize payload capacity.
4. Maintenance: The aircraft should be easy to maintain and repair over time.
5. Financial Cost: Budget constraints were less critical for this competition, so this factor had a lower priority in the analysis.

The AHP results are presented in Table 6.2. Based on the AHP analysis, strength and weight were identified as the most critical factors. Using the results, QFD was conducted for material selection. Before QFD, various combinations of carbon fiber, fiberglass, and aramid were fabricated via vacuum bagging for strength and weight testing.

Table 6.2: AHP Result of composite materials

Significantly More Important 5 ~ Same Importance 1 Criteria	Manufacturability	Strength	Weight	Maintenance	Financial Cost	Priority
Manufacturability	1.00	0.33	0.33	1.00	4.00	0.132
Strength	3.00	1.00	0.50	5.00	5.00	0.320
Weight	3.00	2.00	1.00	4.00	4.00	0.387
Maintenance	1.00	0.20	0.25	1.00	4.00	0.112
Financial Cost	0.25	0.20	0.25	0.25	1.00	0.049



Num	Composite
1	1 × Carbon sheet + 1 × Fiber Glass sheet
2	1 × Aramid sheet + 1 × Fiber Glass sheet
3	2 × Carbon sheet
4	1 × Carbon sheet
5	1 × Fiber Glass sheet
6	1 × Carbon sheet + 1 × Aramid sheet
7	2 × Fiber Glass sheet

Figure 6.2: Specimen Test

The QFD weights were calculated in a similar fashion to the analyses above. Based on the QFD analysis, we decided to use a combination of one sheet of carbon fiber and one sheet of glass fiber for the aircraft skin. $\mathrm{PA-CF}$, a carbon-infused filament, was selected for the motor mount, ribs, and bulkheads. The QFD results are presented in Table 6.3.

Table 6.3: QFD Result of composite materials

Correlation Key		Material	1 × Carbon sheet	1 × Carbon sheet + 1 × Fiber Glass sheet	2 × Carbon sheet	1 × Aramid sheet + 1 × Fiber Glass sheet	LW PLA	PA-CF	Balsa + Plywood
●	Strong (9)								
○	Moderate (3)								
△	Weak (1)								
	None (0)								
Relation	AHP Weight	AHP Criteria							
↑	0.1319	Manufacturability	△				●	●	●
↑	0.3202	Strength		○	●	△		○	
↓	0.3865	Weight	●	●		○	○	○	
↑	0.1124	Maintenance	○				●	●	
↓	0.0489	Financial Cost	△	△					●
		QFD Weight	4.0	**4.5**	2.9	1.5	3.4	4.3	2.0

6.3 | Manufacturing Overview

6.3.1 | Composite Manufacturing of Fuselage

The carbon composite manufacturing process follows these steps:

1. Apply wax and release agent to the mold surface.
2. Attach the perforated film.
3. Laminate carbon fibers and epoxy resin.
4. Apply peel ply and bleeder layers.
5. Perform vacuum operations using a bagging film.
6. Peel out the finished part, and cut it according to design.

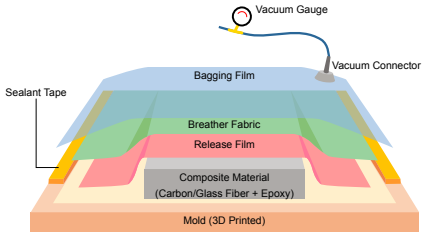


Figure 6.3: FRP Process Sequence

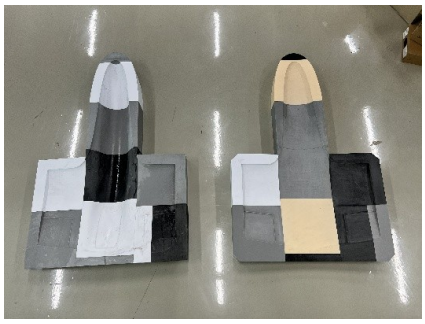


Figure 6.4: Fuselage mold

For the fuselage fabrication, separate molds were prepared for the upper and lower fuselage shells. The lower fuselage was laminated using two sheets of Carbon 3K twill. Resoltech 1050 resin and 1056S hardener were selected for their efficient working, reaction, and curing times, ensuring a controlled fabrication process.

Lamination was optimized for each component. The lower fuselage, requiring high strength for payload support and landing forces, used two sheets of carbon fiber. The upper fuselage had lower structural demands. To reduce weight while maintaining integrity, Carbon 3K twill and Micro Fiber Glass were used.



Figure 6.5: Fuselage Vacuum Operations



Figure 6.6: Fuselage Skin & Bulkhead

Excessive epoxy resin between carbon fibers can reduce stiffness. To prevent this, a vacuum pump removed excess resin, reducing weight and enhancing stiffness. To complete the fuselage's monocoque structure, bulkheads are designed to maintain maximum rigidity in conjunction with the carbon skin. The bulkheads are 3D printed using $\mathrm{PA-CF}$. Two 1×6 mm carbon battens connect them, minimizing weight while optimizing stiffness against vertical forces. In addition, battery plate and inner ribs are added to provide further support and ensure accurate assembly. The upper and lower fuselage skins are bonded using micro-fiberglass and epoxy to provide maximum stiffness while multiple printed formers ensure that the fuselage retains its shape. Note that the canopy is later cut out of the fuselage during manufacturing.



Figure 6.7: Assembling Parts



Figure 6.8: Assembling Parts with Band Clamp

The main fuselage and tail must ensure adequate structural rigidity while offering ease of assembly. Band clamps were used to connect the carbon pipes of the main fuselage and the tail to meet these design requirements.

6.3.2 | Composite Manufacturing of Wings and Empennage

During the pilot production batch, the wing root section was fabricated using three sheets of Carbon 3K twill, while two sheets were used for the wing tip. However, the issue of excessive stiffness and increased weight persisted. Therefore, from the second batch, both upper and lower wing skins were constructed from a single sheet of carbon and a single sheet of glass fiber impregnated with epoxy to optimize weight while maintaining structural integrity.



Figure 6.9: First Test Production Batch Main Wing Mold



Figure 6.10: First Test Production Batch Vacuum Operation with Root Press mold

To imprint control surface outlines on the carbon laminate, separately fabricated guideline parts were used as cutting references after the primary structure was manufactured. Figures 6.11 and 6.12 show the main wings of the first and second prototypes. The different wing shape in the second prototype is due to the application of FORCAST optimization results. The main wing and empennage were manufactured using the same process as the fuselage.

Carbon rod fittings with smaller rods are inserted into larger bonded tubes to connect the main wing sections and the fuselage together. Quick release mounts are selected to fix the main wing and fuselage spanwise to make installation and removal convenient. The mounts mainly have to resist spanwise forces, since most of the bending moments are imparted to the carbon rods. The wing dihedral spars meet at a dihedral angle of 5° in the middle and are securely joined through forging.

The tail wing attaches via booms fitted to the fuselage and tail section. Carbon paste increases friction, and pipe clamps secure the connection. Additionally, the vertical stabilizers can be separated from the rest of the assembly for easier transportation.



Figure 6.11: Pilot Production Batch

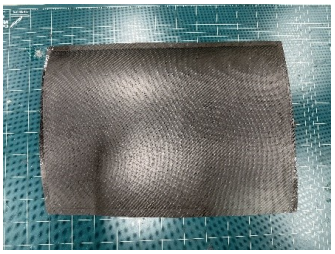


Figure 6.12: Second Pilot Batch

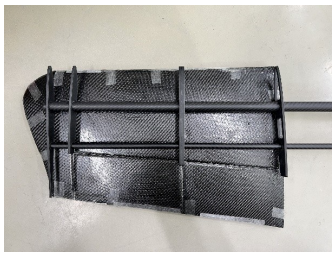


Figure 6.13: Structure of Main wing

The internal structure of the main wing consists of spars and ribs. The ribs are fabricated using a 3D printer with PA-CF filaments, while the spars are constructed using carbon pipes.

6.3.3 | Composite Manufacturing of Landing Gear

The landing gear, designed to distribute and withstand total load, was made from 12 sheets of Carbon 3K twill. These sheets were laminated over a basic mold, then trimmed with a grinder to remove excess material.



Figure 6.14: **Landing gear Pilot production Batch**

6.3.4 | Manufacturing of the X-1 Vehicle

For the X-1 vehicle, simplifying the manufacturing process was a top priority, as rapid prototyping was required to test its design and automatic flight controls. Therefore, the X-1 prototype was fully 3D printed using LW-PLA. To improve manufacturability, the fuselage, wing, and winglet were printed separately and assembled later.

However, the weak bonding between the layers led to frequent fractures during flight tests. Furthermore, since the X-1 weighed approximately 230 g even using LW-PLA, further weight reduction was necessary. To address these issues, balsa wood was chosen as the primary material for the X-1. The basic shape was maintained, while all components including the spar and ribs were replaced with balsa wood. The fuselage was hollowed out to accommodate the flight controller and LED circuit. To minimize weight, the exterior was covered with Monokote film, ensuring the design remained lightweight and functional for flight tests.

6.4 | Summary

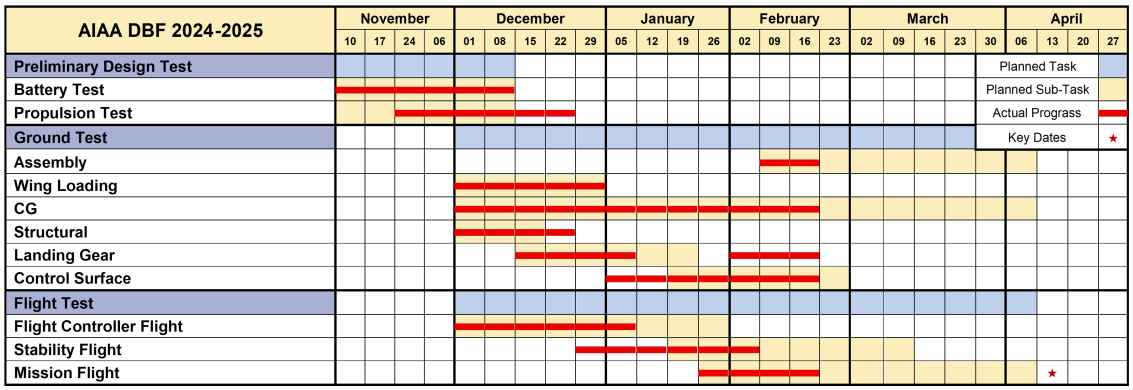
Table 6.4: Summary of Aircraft Manufacturing Processes

Manufacturing			
Aircraft component	Parts	Materials	Processes
Fuselage	Skin	Carbon 3K (plain 5.9 oz)	CFRP GFRP Hybrid Manufacturing
		Micro Fiberglass	
		Epoxy	
	Bulkhead	PA-CF filament	3D printed
	Spar	Carbon pipe	-
	Ribs	PA-CF filament	3D printed
Propeller Assembly	Purchased including propeller, spinner, and attachment hardware (APC)	GFRP	-
Motor	Purchased including attachment harware and mounting plate	PA-CF filament	3D printed
	Mount	PA-CF filament	3D printed
Battery	Purchased (SMC)	-	-
Landing Gear	Nose Gear	Purchased	Purchased
	Body Gear	Carbon 3K (plain 5.9 oz)	CFRP GFRP Hybrid Manufacturing
		Epoxy	
Vertical Tail	Skin	Carbon 3K (plain 5.9 oz)	CFRP GFRP Hybrid Manufacturing
		Micro Fiberglass	
		Epoxy	
	Ribs	PA-CF filament	3D printed
	Spar	Carbon pipe	-
	Rudder	Carbon 3K (plain 5.9 oz)	CFRP GFRP Hybrid Manufacturing
		Micro Fiberglass	
		Urethane Foam	
		Epoxy	
Horizontal Tail	Skin	Carbon 3K (plain 5.9 oz)	CFRP GFRP Hybrid Manufacturing
		Micro Fiberglass	
		Epoxy	
	Ribs	PA-CF filament	3D printed
	Spar	Carbon pipe	-
	Elevator	Carbon 3K (plain 5.9 oz)	CFRP GFRP Hybrid Manufacturing
		Micro Fiberglass	
		Urethane Foam	
		Epoxy	
	Mount	PA-CF filament	3D printed
Main Wings	Skin	Carbon 3K (plain 5.9 oz)	CFRP GFRP Hybrid Manufacturing
		Micro Fiberglass	
		Epoxy	
	Ribs	PA-CF filament	3D printed
	Spar	Carbon pipe	3D printed
	Flap	Urethane Foam	CFRP GFRP Hybrid Manufacturing
		Carbon 3K (plain 5.9 oz)	
		Epoxy	
		Micro Fiberglass	
	Aileron	Urethane Foam	CFRP GFRP Hybrid Manufacturing
		Carbon 3K (plain 5.9 oz)	
		Epoxy	
		Micro Fiberglass	

7 | Testing

The test was carried out according to the proposal, divided into three stages: preliminary design test, ground test, and flight test. The preliminary design test included a propulsion test to evaluate and determine the optimal combination of motor, propeller, and battery. The ground test was performed to assemble the components and verify their individual performance. Finally, the flight test was carried out to assess the overall performance of the system in real flight conditions.

Table 7.1: Test Schedule



7.1 | Test Objectives

Table 7.2: Test Objectives

Test Type		Aircraft	Objective
Preliminary Design Test	Battery Test	Main	Check the voltage drop according to the discharge current.
	Propulsion Test	Main	Evaluate different combinations of motors and propellers.
Ground Test	Assembly	Both	Simulate each ground missons and minimize assembly time.
	Wing Loading	Main	Lift at a single point on each wingtip to verify sufficient wing strength.
	CG	Both	Verify the CG point whether it is on the acceptable region.
	Structural	Main	Ensure aircraft can withstand loadings during flight or ground mission.
	Landing Gear	Main	Evaluate strenght of the landng gear.
	Control Surface	Both	Check whether the servo motor can withstand the hinge moment.
Flight Test	Flight Controller	X-1	Check whether the X-1 can perform autonomous flight.
	Stability	Both	Ensure the aircraft is stable during flight.
	Mission Flight	Both	Ensure whether both aircrafts can perform all missions properly.

7.2 | Preliminary Design Test

Battery Test

The battery test was conducted to evaluate the performance of the selected battery and provide discharge data for FORCAST. The voltage was measured by discharging the battery at constant currents of 30 A and 80 A via a motor. The data obtained were then used for propulsion analysis. This process is detailed in Section 4.5.

Propulsion Test

The propulsion test was conducted to evaluate three final motor-propeller combinations selected during the preliminary design phase. The Mayatech MT10 Pro test bench was used for the test. The power and thrust of the motor alongside temperatures were recorded to assess the performance and stability of the propulsion system. Additionally, the motor mount intended for the aircraft was installed to assess its durability.

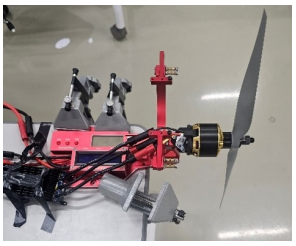


Figure 7.1: Propulsion Test

7.3 | Ground Test

Assembly

An assembly test was conducted to install the battery pack, the X-1 including the launch system, and fuel tanks within five minutes. The process was documented by recording the test and manually noting the procedures. The assembly and disassembly processes were repeated multiple times to prepare for the ground test. Simplifying the procedures to reduce the assembly time was crucial. Additionally, for the X-1, the parameters were adjusted to arm immediately after calibration, ensuring reliability.

Wing Loading

The wing loading test was conducted to ensure that the wing had sufficient structural strength. The test was performed by lifting the wings as in the actual competition, and confirmed structural integrity.

CG

Due to the selection of a high thrust-to-weight ratio motor to maximize mission scores, the CG point was determined to maintain a static margin of 20%. The landing gear positions were determined to ensure that the desired CG point falls in an 6:1 ratio between the nose gear and body gear. The measured data from a custom-built CG and weight measuring tool was used to find and help set the CG point.



Figure 7.2: Aircraft mounted on CG Measuring Tool

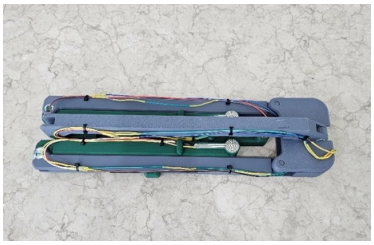


Figure 7.3: Folded up CG Measuring Tool

Structural

The structural test was conducted to determine whether the materials used were appropriate by testing how much they deform under stress. The test involved manufacturing composite samples made of various materials (e.g. Carbon Fiber) and performing stiffness tests. This process is described in Section 6.2.4.

Landing Gear

The landing gear test was conducted to ensure its durability and proper functionality. Different loads were applied to the landing gears to evaluate its load-bearing capacity and the extent of deformation it could withstand upon impact.

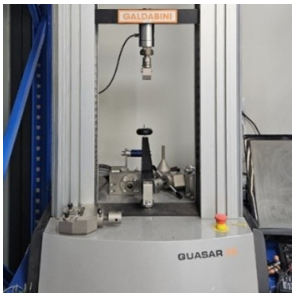


Figure 7.4: **Landing Gear Bending Test**

Control Surface

By analyzing the geometry of the aircraft and flight conditions, the aerodynamic force applied to the control surface was estimated to select appropriate servos. To model the landing condition where the flap undergoes the highest moment, the following conditions were assumed: a stall angle of 12.5°, landing speed of 20 m/s, and a flap angle of 45°. Analysis using OpenVSP determined a moment coefficient $C_m$of -0.25, and the maximum force the servo must withstand was calculated as 26.42 N/cm.

$$M = \frac{1}{2}\rho V^2 \cdot C_m \cdot S \cdot c \tag{7.1}$$

To ensure a factor of safety greater than 2, a servo capable of providing 6 kg/cm torque at 7.4 V was selected. Load test on the actual servo confirmed that it could withstand a maximum load of 5.6 kg/cm, with a factor of safety of 2.23.

7.4 | Flight Tests

Stable Flight

A stability test was carried out to verify whether the X-1 test vehicle could fly properly without flight controls. A 3D-printed X-1 prototype was towed and released by a drone to assess its glide capability, stability, and sink rate. The sink rate was used as a reference indicator when adjusting control parameters.



Figure 7.5: **X-1 flight test**

Flight Controller Flight

The X-1 was tested again with a control surface to evaluate the automatic flight controls before testing with the main aircraft. The test was conducted similarly to the stability test above.

Mission Flight

The mission flight test was conducted to assess whether the aircraft and X-1 could successfully perform mission 2 and 3, and to identify potential improvements during flight. A flight controller was mounted in the main aircraft during tests for obtaining statistics. The results were applied to the pilot manual to maximize scores using the same aircraft.

7.5 | Checklists

Table 7.3: Checklists

General					
Battery check		Fuel tank installation		Hatch, Canopy check	
Aera clear check		X-1 installation		Sensors calibration	
Main aircraft				**X-1**	
Control surface check		Controller connection		X-1 Mission upload	
CG check		Kill switch check		Autopilot check	
Motor check		Release check		LED check	
After Flight					
Throttle cut		Battery unplugging		Arming switch	

8 | Performance Results

8.1 | Propulsion

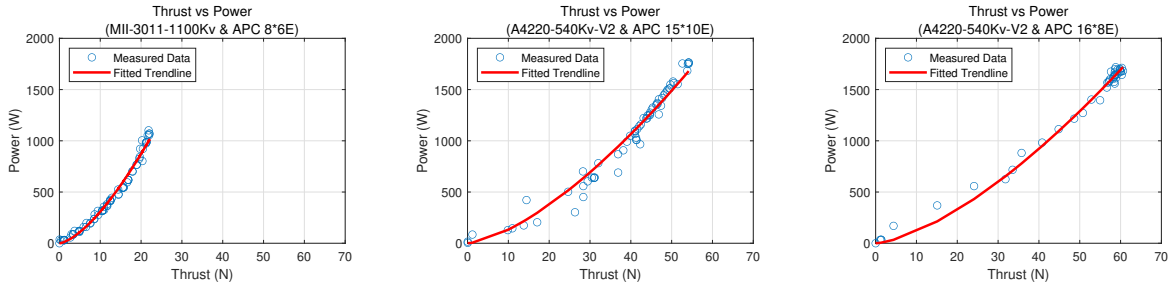


Figure 8.1: Thrust vs Power Graphs

The power-thrust plots in Figure 8.1 illustrate three different motor-propeller combinations. Based on the analysis in Section 4.5.2, the third combination was expected to produce the highest thrust at low speeds. To validate these results, a ground test was conducted with all three types under static conditions. The results confirmed that the third combination generated the highest thrust for the same power input, as expected.

8.2 | Ground Mission

The Ground Mission is a timed mission of the X-1 Flight Test Program. There are two key approaches to achieving a full score: optimizing the training process and designing efficient pylons to reduce mission time. Table 8.1 presents the results of the ground test. The first attempt was conducted using the initial prototype of the release mechanism, while a simplified and improved version was implemented from the second attempt onward. As a result, the ground mission time was significantly reduced.

Table 8.1: Ground Mission

Successful: O Failure: X	1	2	3	Mean
Pylon (s)	25	23	21	23
Fuel tank, X-1 (s)	21	13	11	15
Flight control	O	O	O	-
Release	O	O	O	-
LED	X	O	O	-
Total time (s)	46	36	33	38

8.3 | Test Flight

Figure 8.4, 8.5 show predicted data and test flight results. The test flight aimed to assess flight performance (including turn radius and maneuverability), verify the release mechanism, X-1 landing accuracy, and functionality of LED circuitry.

The test flight, conducted with only the X-1 mounted and no fuel tank, reached a maximum speed of 35 m/s, demonstrating stability and exceptional maneuverability despite strong winds. While the flight path showed slight deviations from the simulation, key metrics such as acceleration and climb rate aligned closely with predictions. The turn radius was initially larger than expected, but in Test 2, after multiple laps, the aircraft achieved a smaller turn radius, aligning more closely with the predicted path.



Figure 8.2: X-1 Release



Figure 8.3: X-1 Flight Path

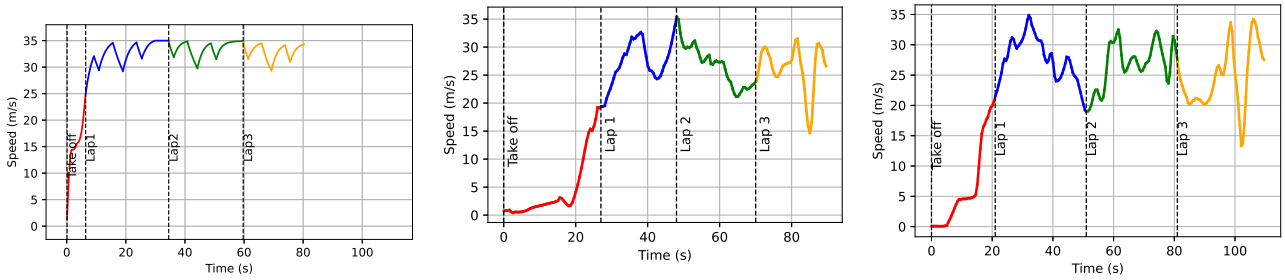


Figure 8.4: Total Speed (left to right: Predicted, Test 1, Test 2)

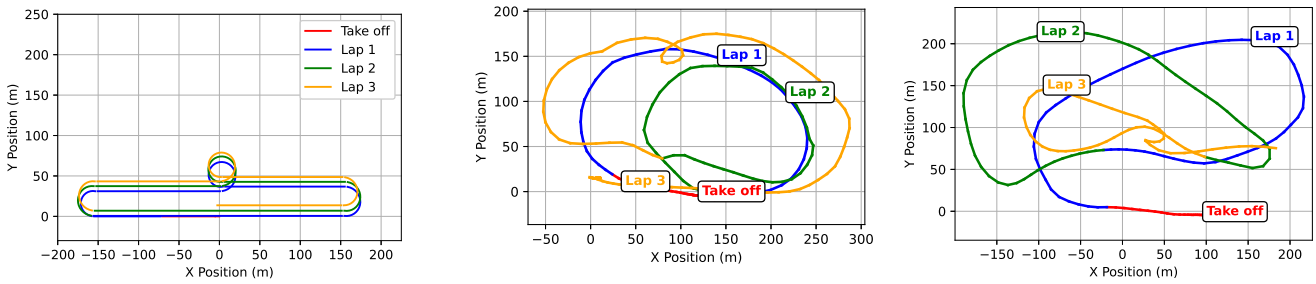


Figure 8.5: Top Down Path (left to right: Predicted, Test 1, Test 2)

After release, the X-1 maintained stability at high speeds, activated its LED, and autonomously loitered at the designated position. When released from 70 meters, it glided for about 20 seconds. The initial release mechanism was too complex for precise timing, so it was simplified and optimized to reduce weight and improve accuracy.

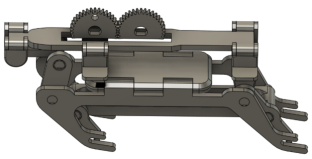


Figure 8.6: Release Mechanism rev.1

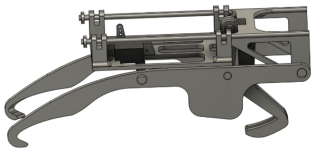


Figure 8.7: Release Mechanism rev.2

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